Topic 4
Sensation and perception: Vision

Overview

- The senses
- Vision – the retina
- Vision – beyond the retina
- Visual perception
- Perceptual organization
- Perception of depth
- Perception of size
- Perceptual constancies

Sandro Del-Prete

Sensation ≠ Perception

Sensation ≠ Perception

- The underside of our brain's right hemisphere → helps us recognize a familiar human face as soon as we detect it
  - Prosopagnozia example

- Human ears are most sensitive to sound frequencies that include human voices, especially a baby's cry

- Frogs, which feed on flying insects, have cells in their eyes that fire only in response to small, dark, moving objects.

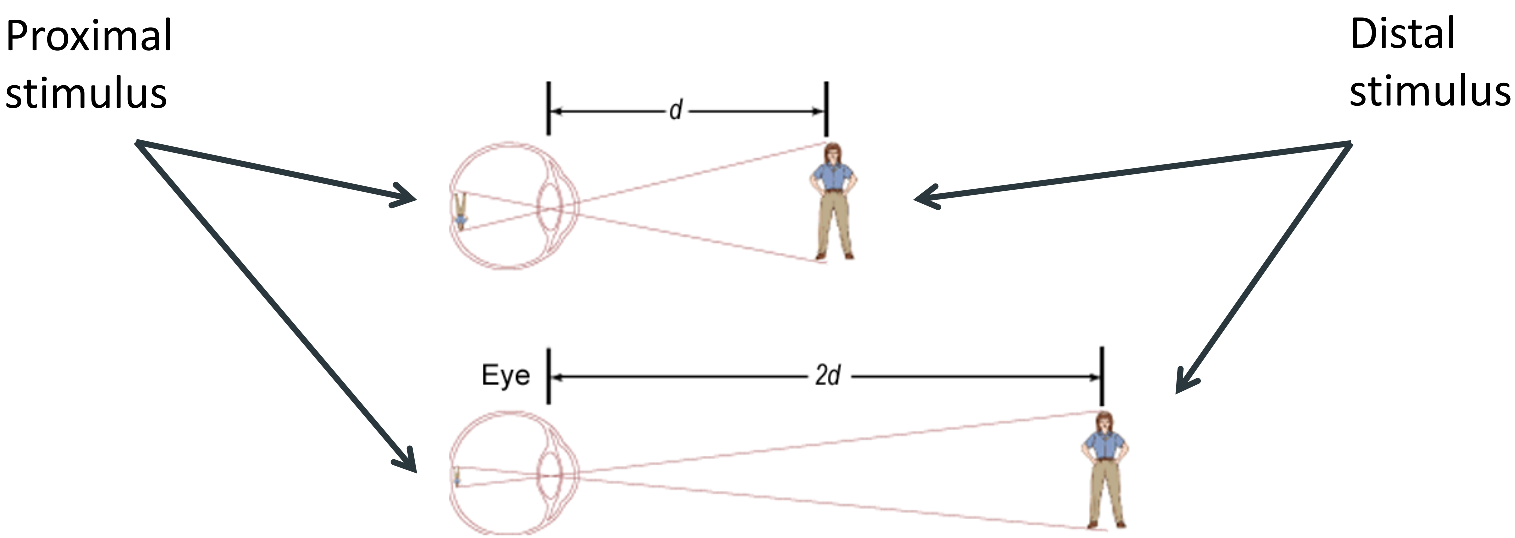
The senses

- **Sensation** - the detection of external stimuli and the transmission of this information to the brain
- **Perception** - the processing, organization, and interpretation of sensory signals

Distal Stimulus → Proximal Stimulus → Transduction → Sensation → Perception

The senses

Are our perceptions a true reflection of reality or do they represent an interpretation of reality?



Thresholds

- We have very restricted awareness:
  - **Absolute threshold**
    - The minimum stimulus energy needed to detect a particular stimulus 50 percent of the time.
      - Can I detect this sound?

Thresholds

- We have very restricted awareness:
  - **Absolute threshold**
    - The minimum stimulus energy needed to detect a particular stimulus 50 percent of the time.
      - Can I detect this sound?
    - **Subliminal**
      - Below absolute threshold
      - If you cannot detect it ***consciously*** at least 50% of the time, it is subliminal for you

Thresholds

- **Difference threshold**
  - Detecting small differences between stimuli
  - The minimum difference between two stimuli required for detection 50 % of the time.
  - We experience the difference threshold as a just noticeable difference

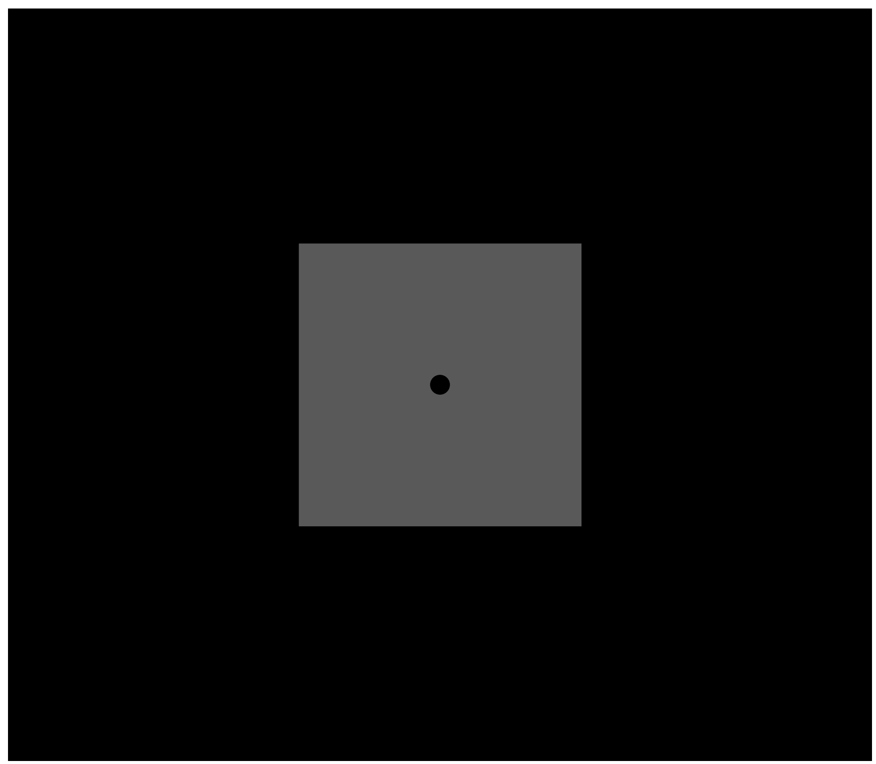
  - **Weber's law:**
    - The more intense the stimulus, the bigger the difference needs to be

The senses

- Vision
- Hearing
- Taste
- Smell
- Skin senses (pressure, temperature, pain)
- Vestibular sense
- Kinesthesia
- Proprioception
- Interoception

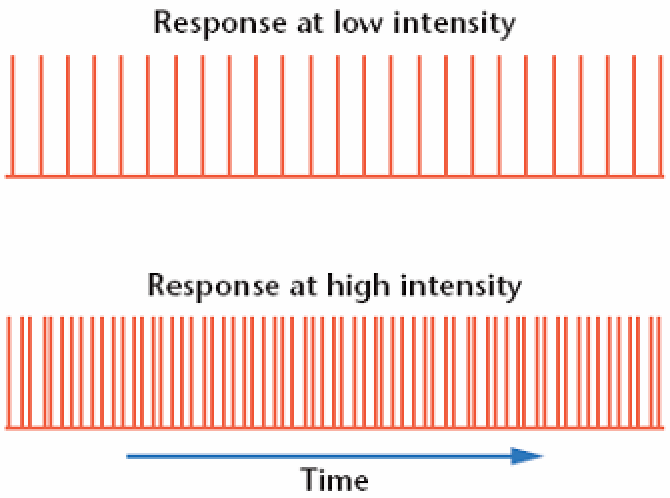
The senses

- Quantitative variations (intensity)



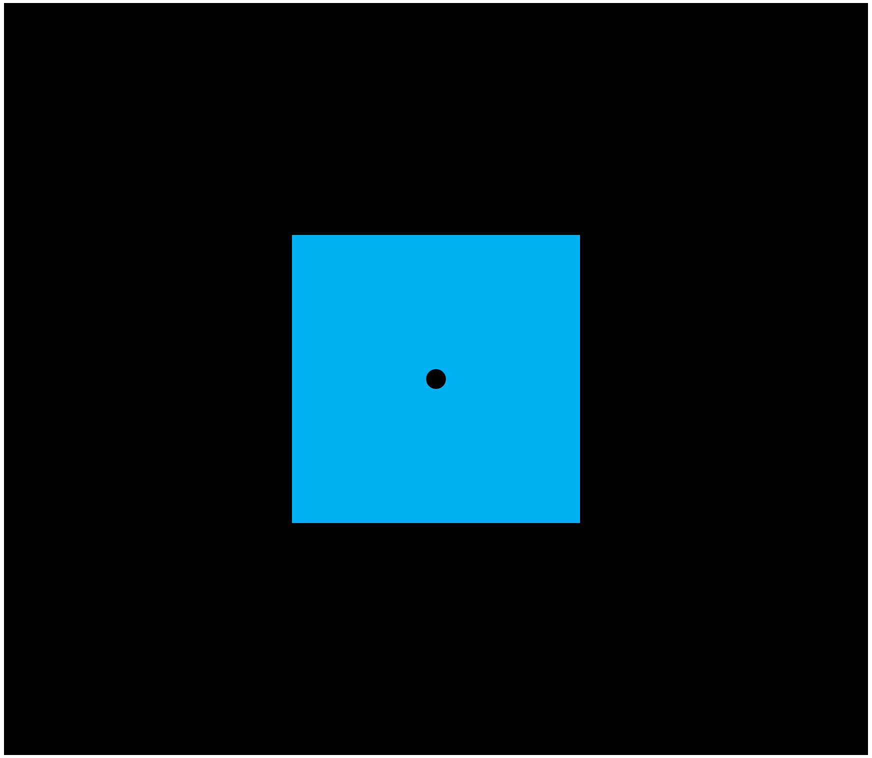
The senses

- Quantitative variations (intensity)
  - Rates of neuron firing
  - Total number of neurons triggered



The senses

- Qualitative variations (sensory quality)



The senses

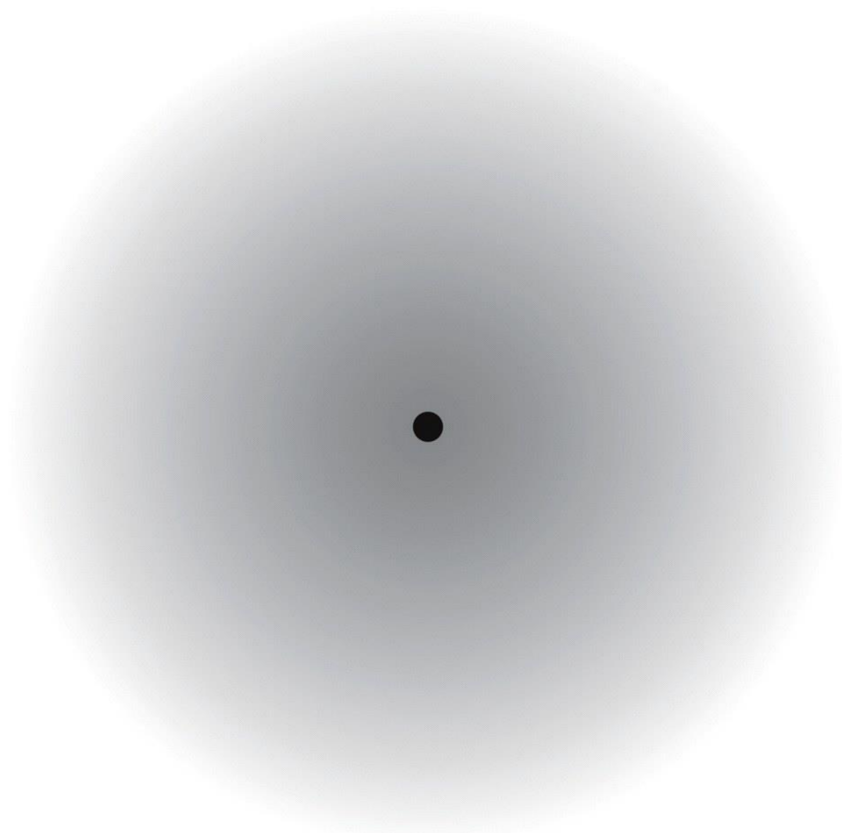
- Qualitative variations (sensory quality)
  - Different sensory qualities are signaled by different neurons
  - Certain sensory qualities arise because of different patterns of activation across a whole set of neurons

The senses

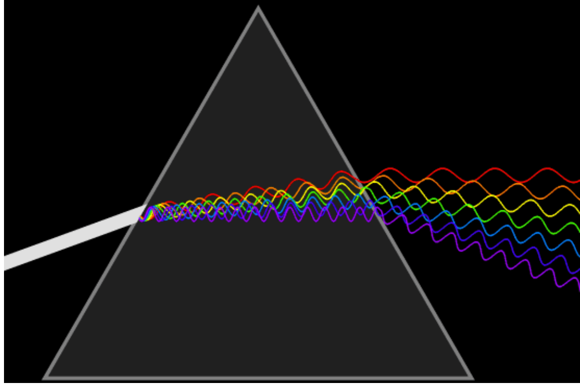
- Sensory Adaptation
  - the tendency to respond less to a stimulus that has been present and unchanging for some time
  - It is at the level of sensory receptors (no response)
  - Benefit:
    - Freedom to focus on informative changes in our environment

The senses

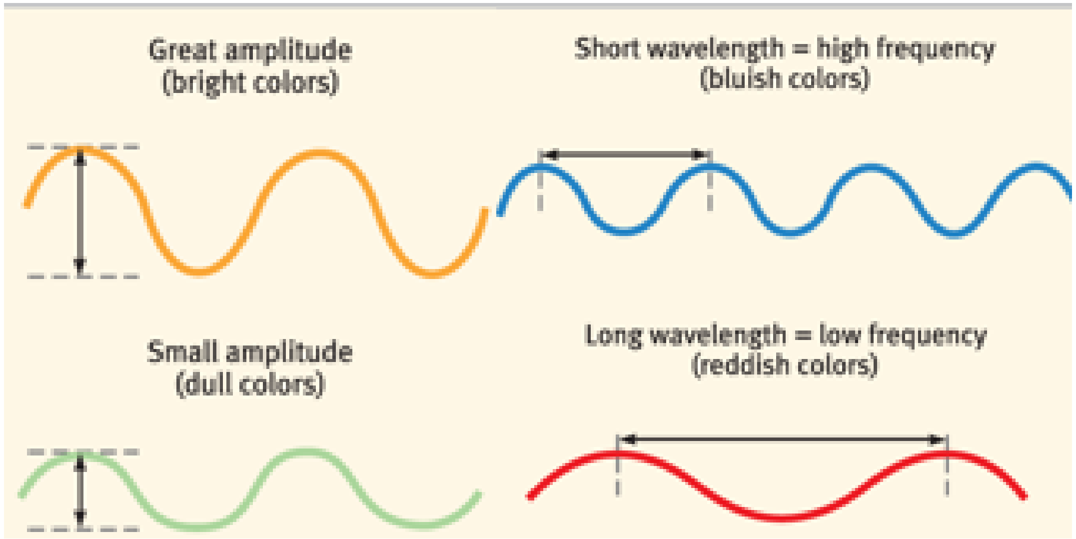
- Adaptation



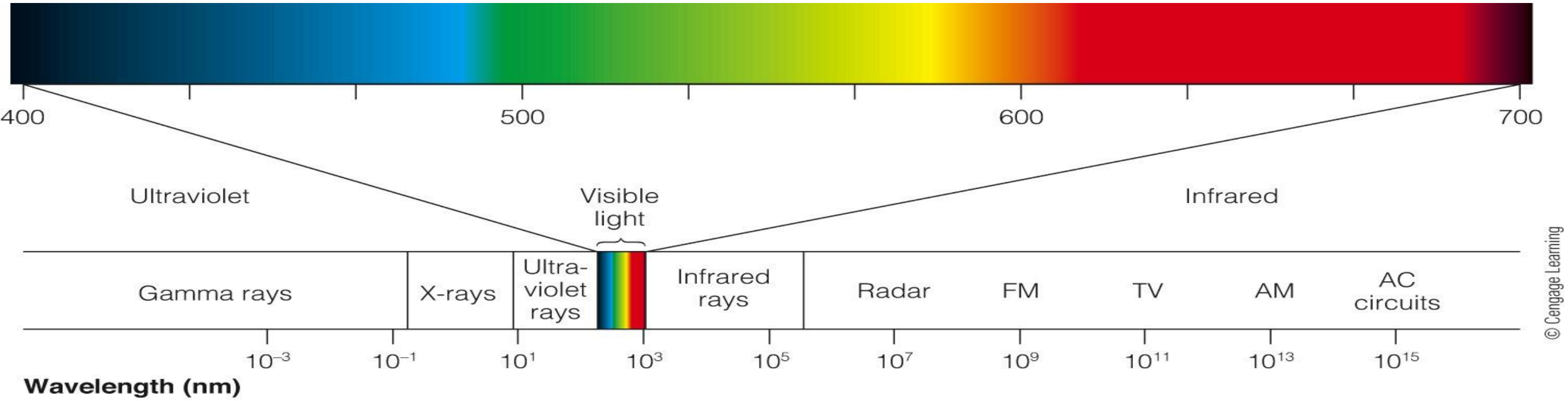
Vision – the retina



- Light (direct or reflected)
  - can vary in intensity



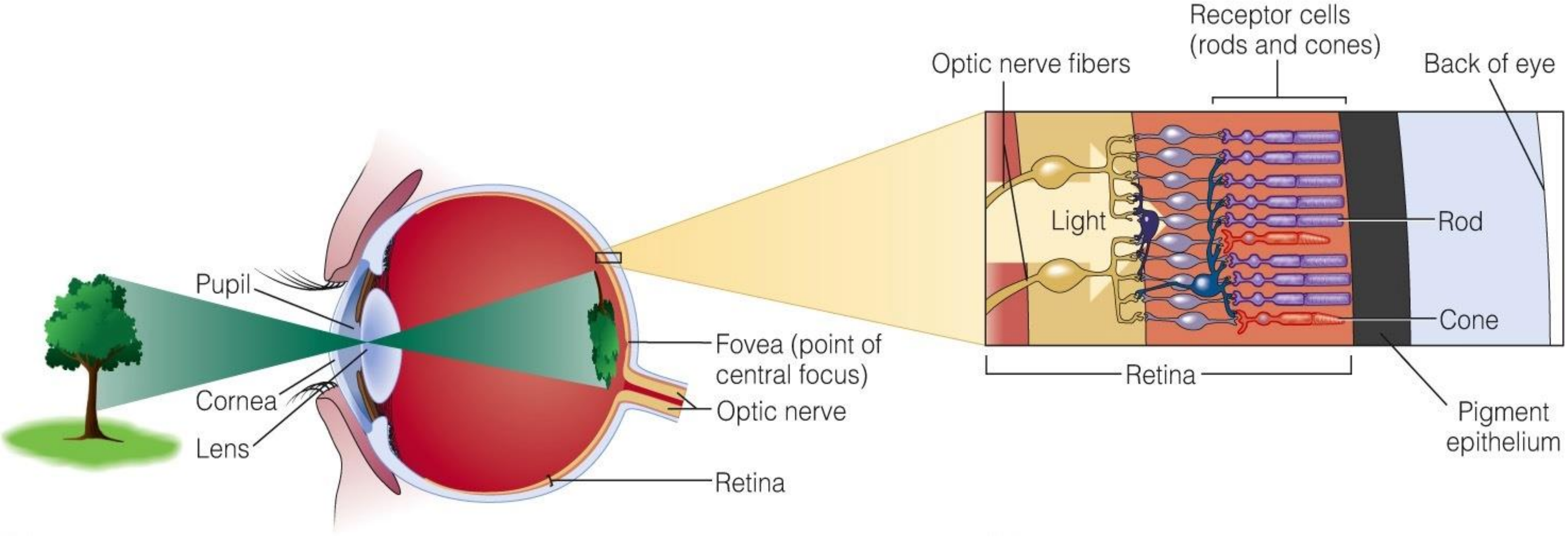
  - can vary in wavelength



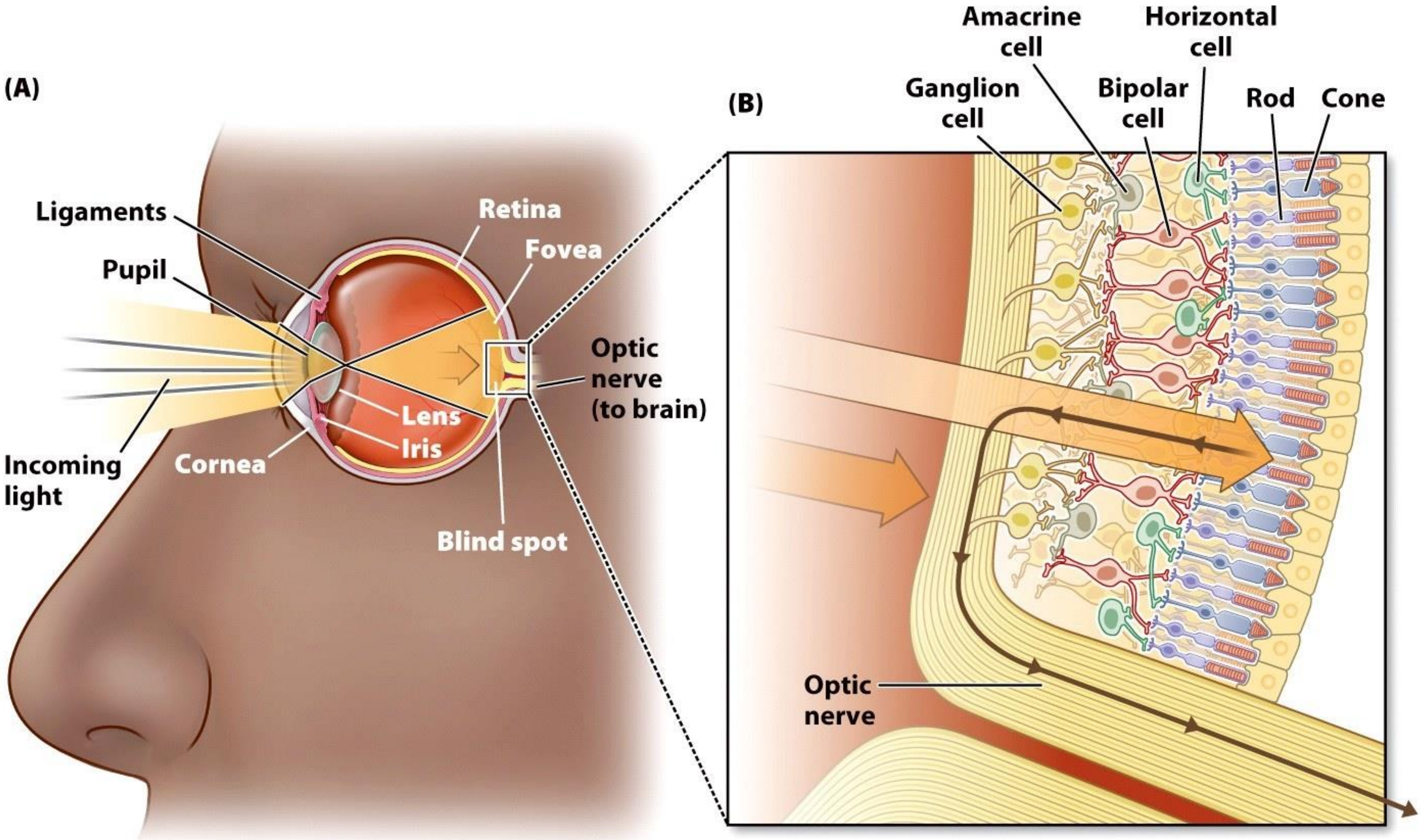
Vision – the retina

Light → Cornea →Pupil, Iris → Lens→Retina→Optic nerve→Brain's visual cortex

Bottom-up processing



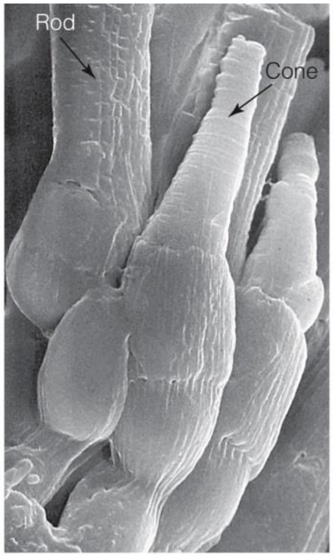
Vision – the retina



Vision – the retina

- Light (direct or reflected)
  - can vary in intensity and wavelength
  - is absorbed by pigments in the receptors (rods and cones) which transduce light energy into electric energy ⟹
  - electric signal is carried to the brain via ganglion cells

Vision – the retina



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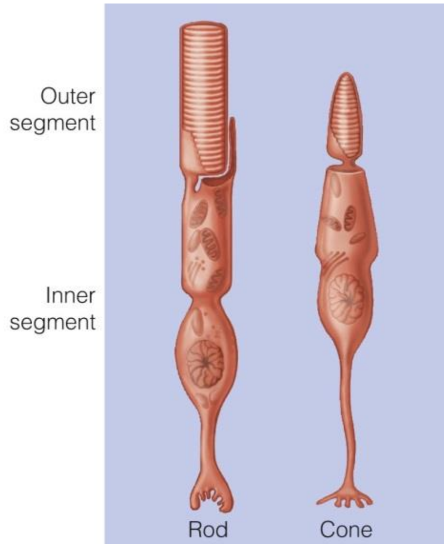

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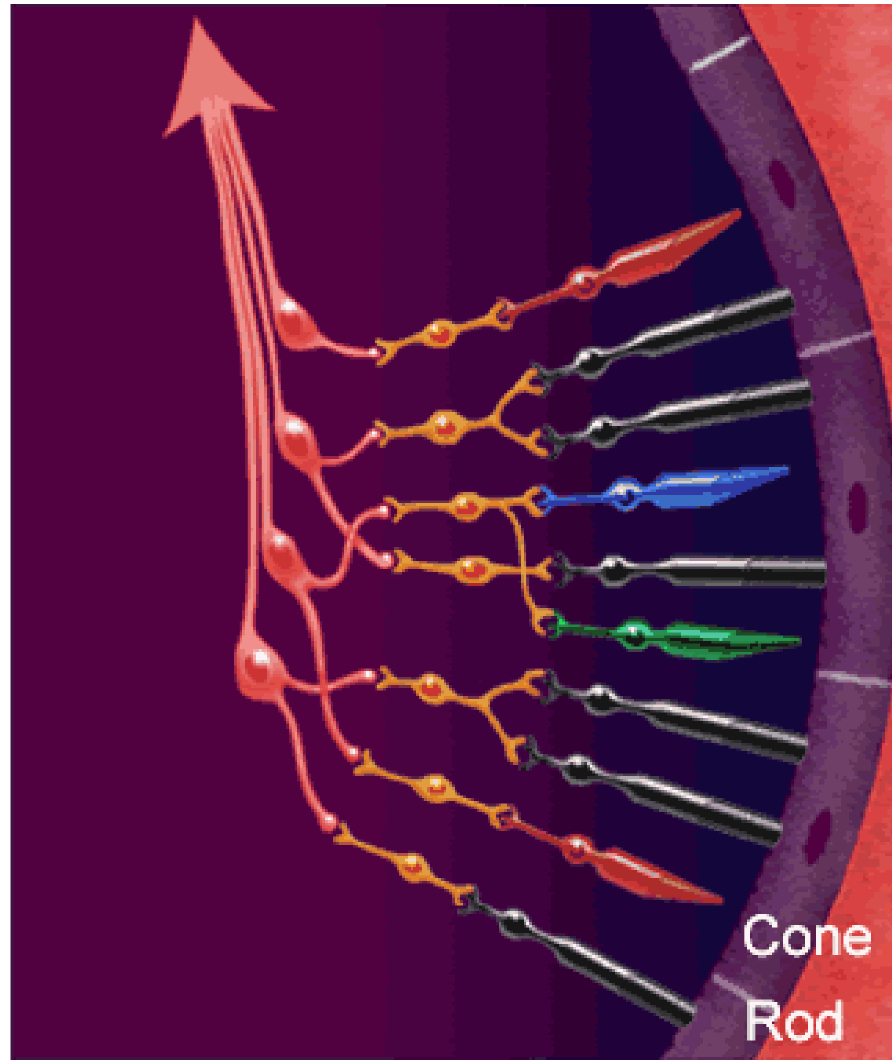
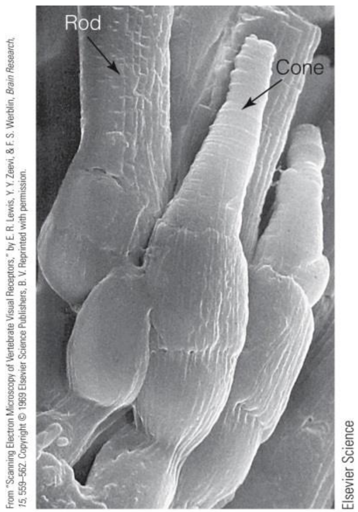
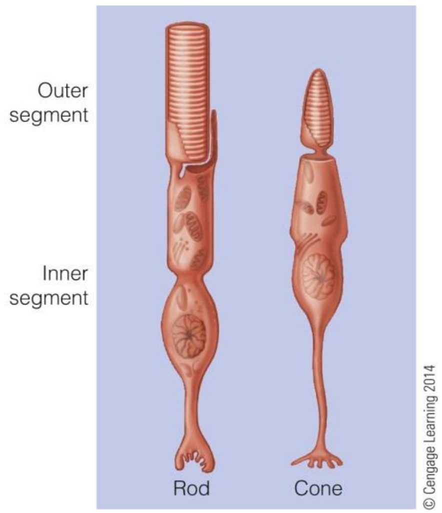


https://storify.com/MariaVita1/opponent-process-theory

Vision – the retina



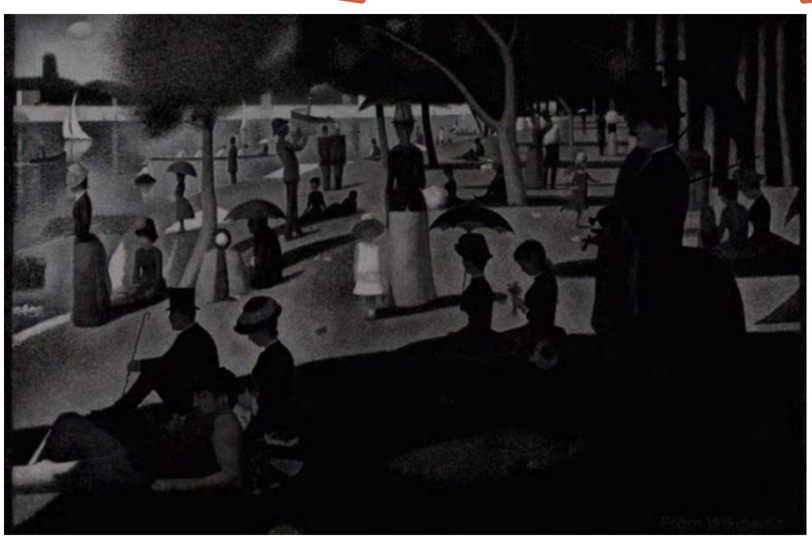
(a)

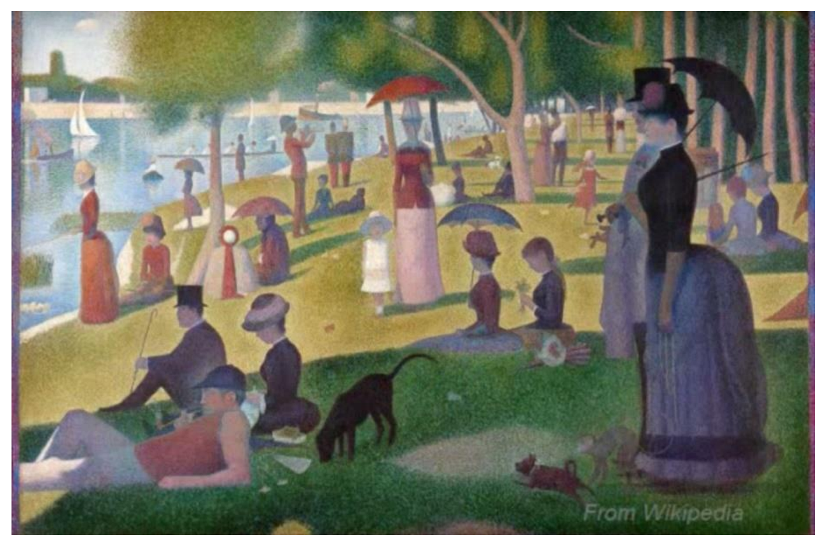


(b)

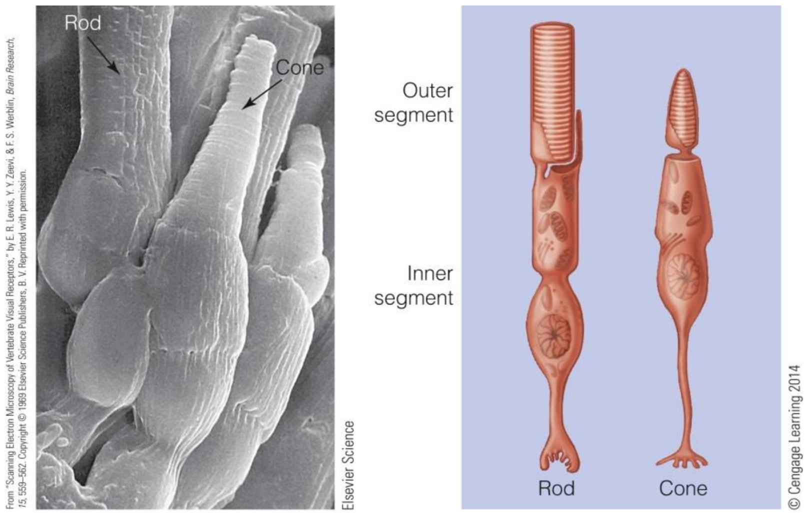
Rods → night vision

Cones → day vision





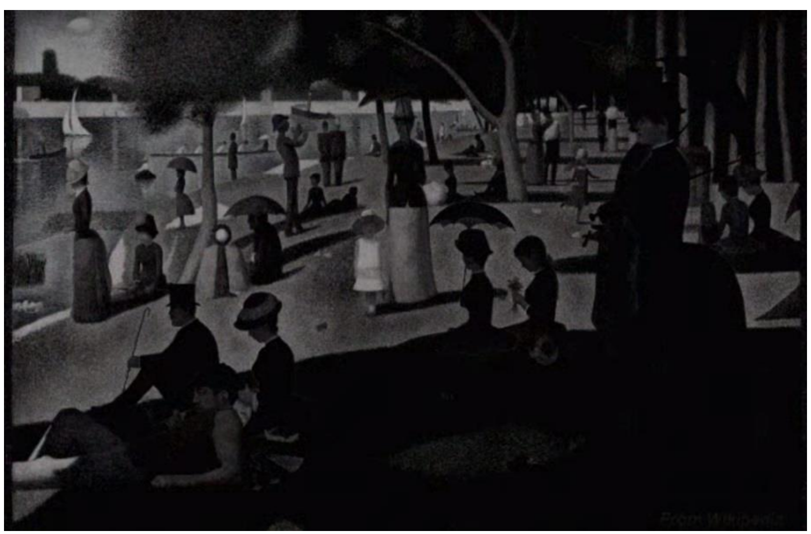
Vision – the retina

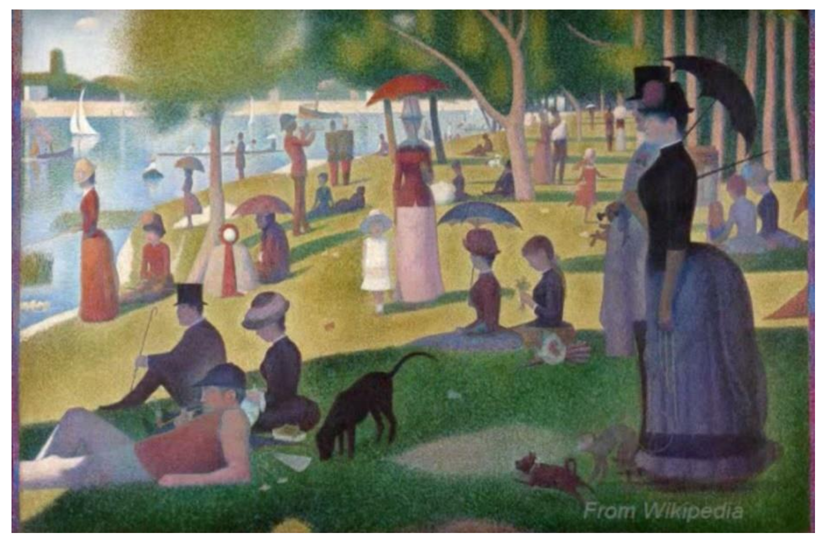


(a) (b)

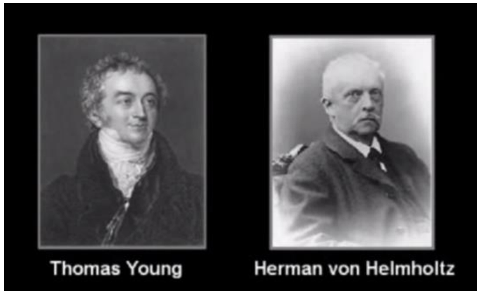
Rods → no color vision

Cones → color vision

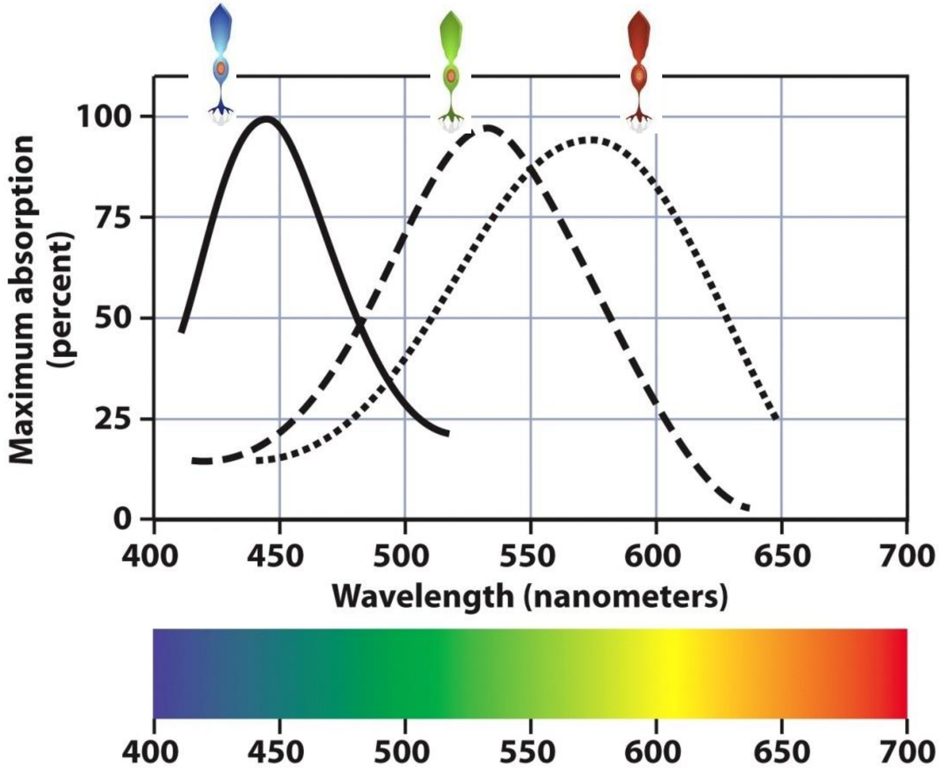


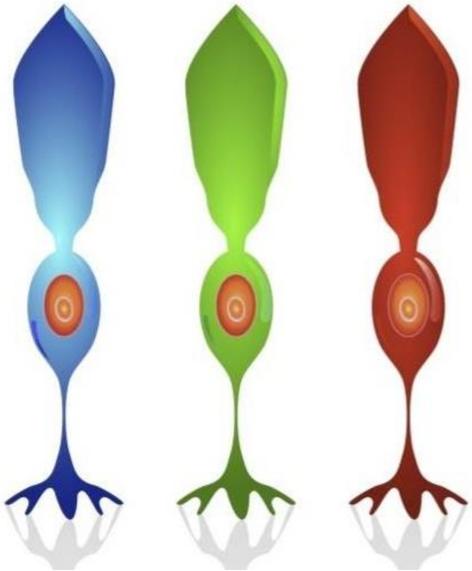


Vision – the retina

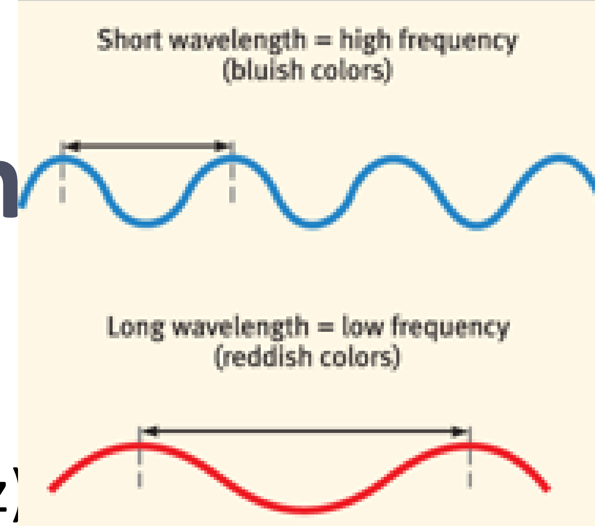


- Color vision
  - Trichromatic Theory (Young-Helmholz)





Vision – the retin



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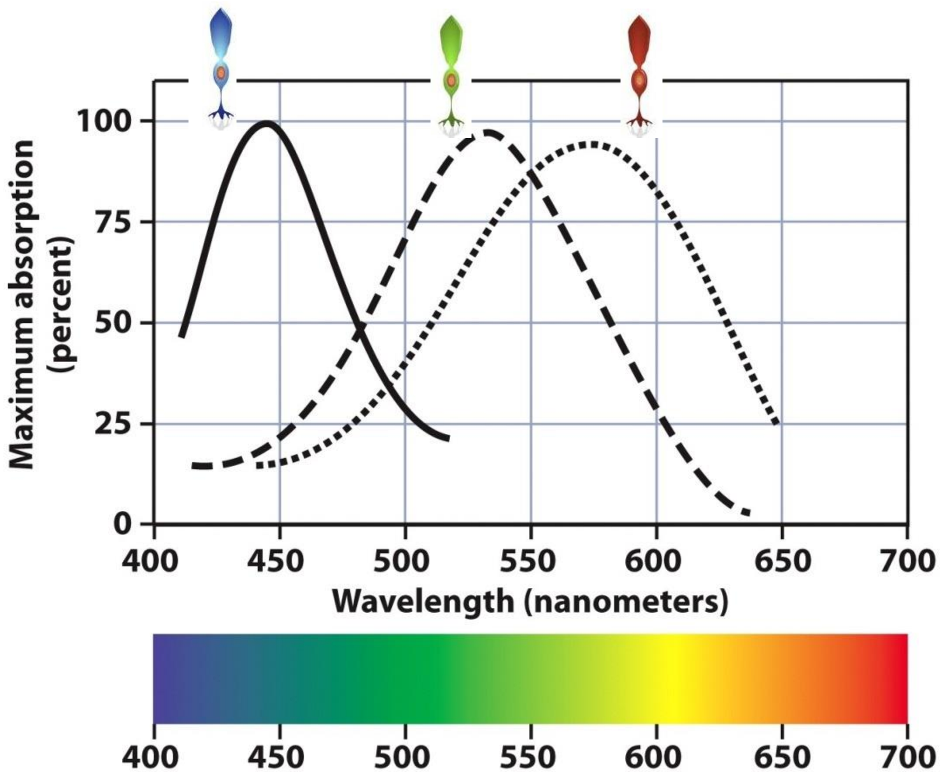
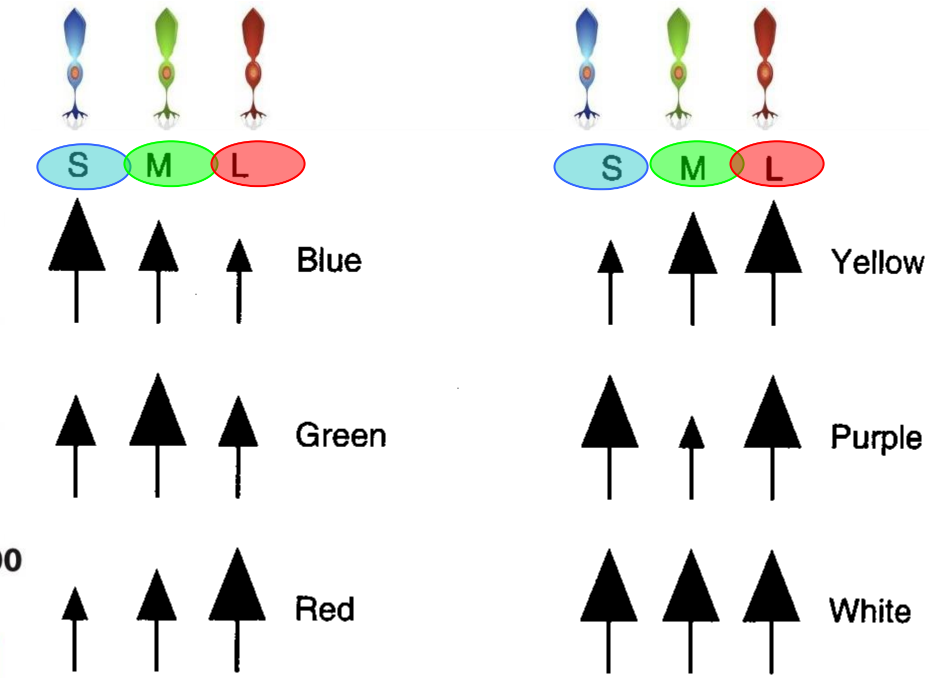
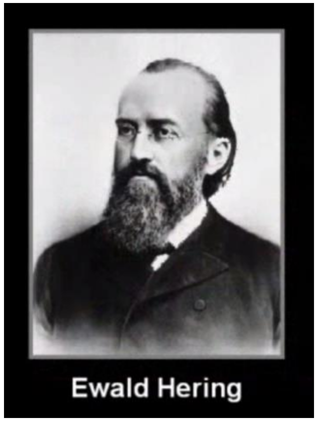
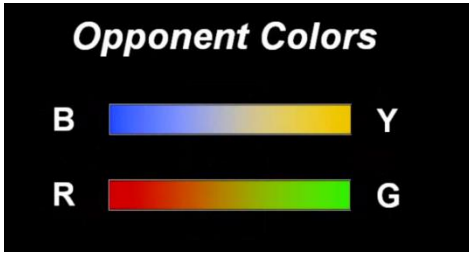




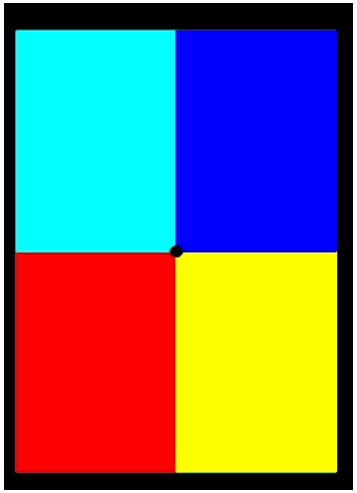
Vision – the retina



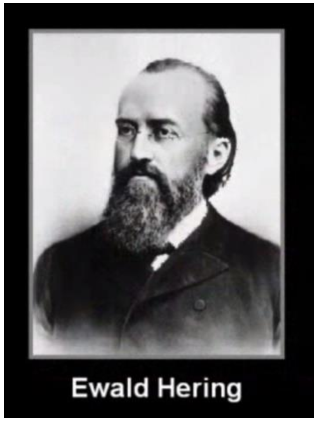
- Color vision
  - Opponent-process theory (Hering)
    - There is no such thing as “blue-yellow” or “red-green”



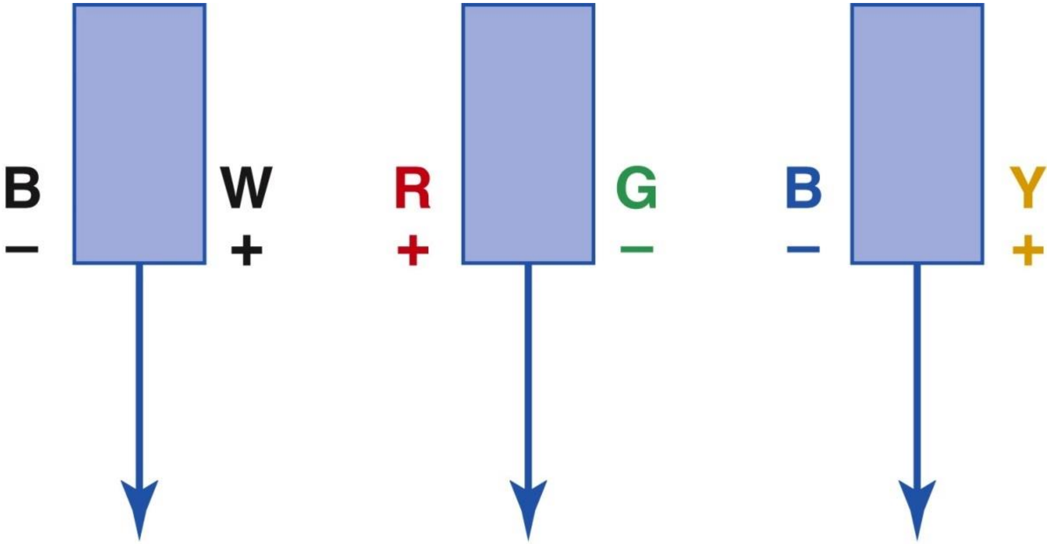
    - Afterimages



Vision – the retina

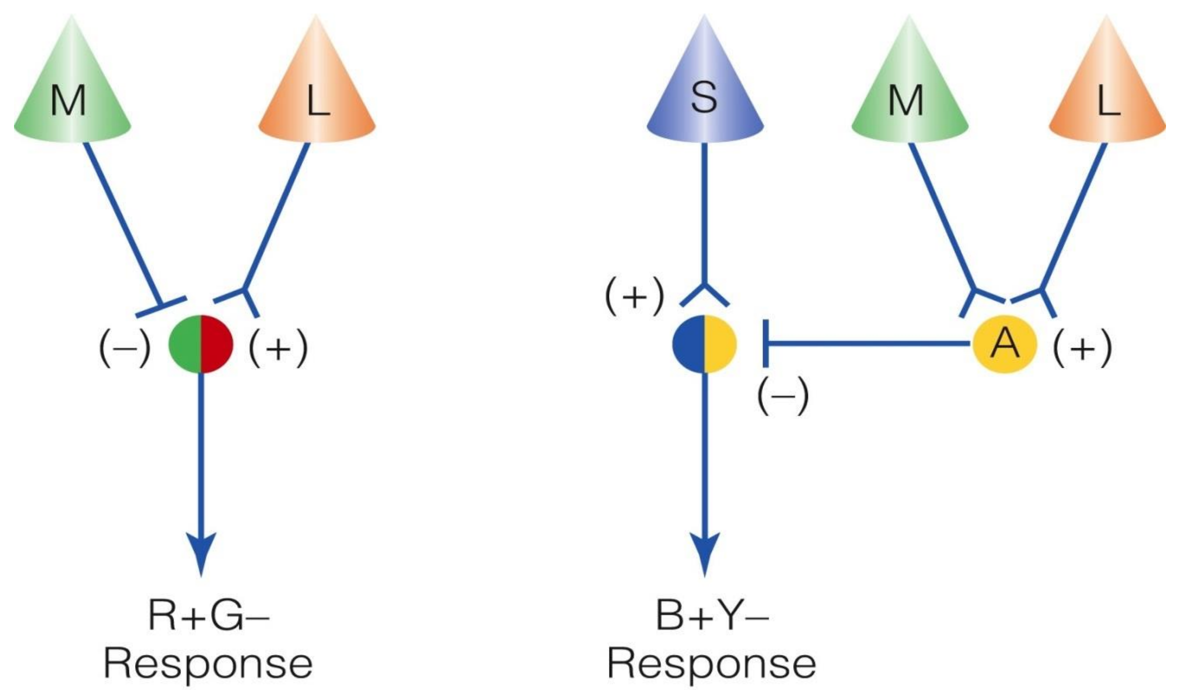


- Color vision
  - Opponent-process theory (Hering)
    - Black-white
    - Red-green
    - Blue-yellow



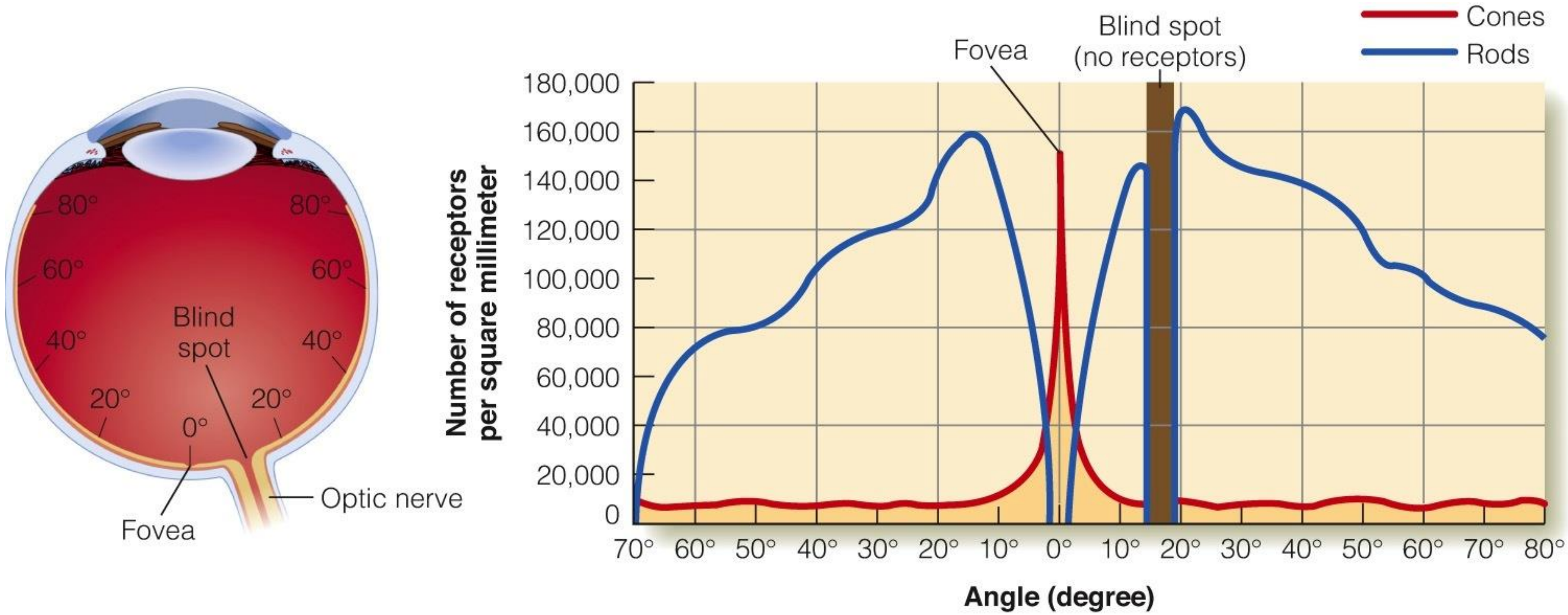
Vision – the retina

- Color vision
  - Trichromatic theory (Young-Helmholtz) and the opponent-process theory (Hering) represent different stages of visual processing



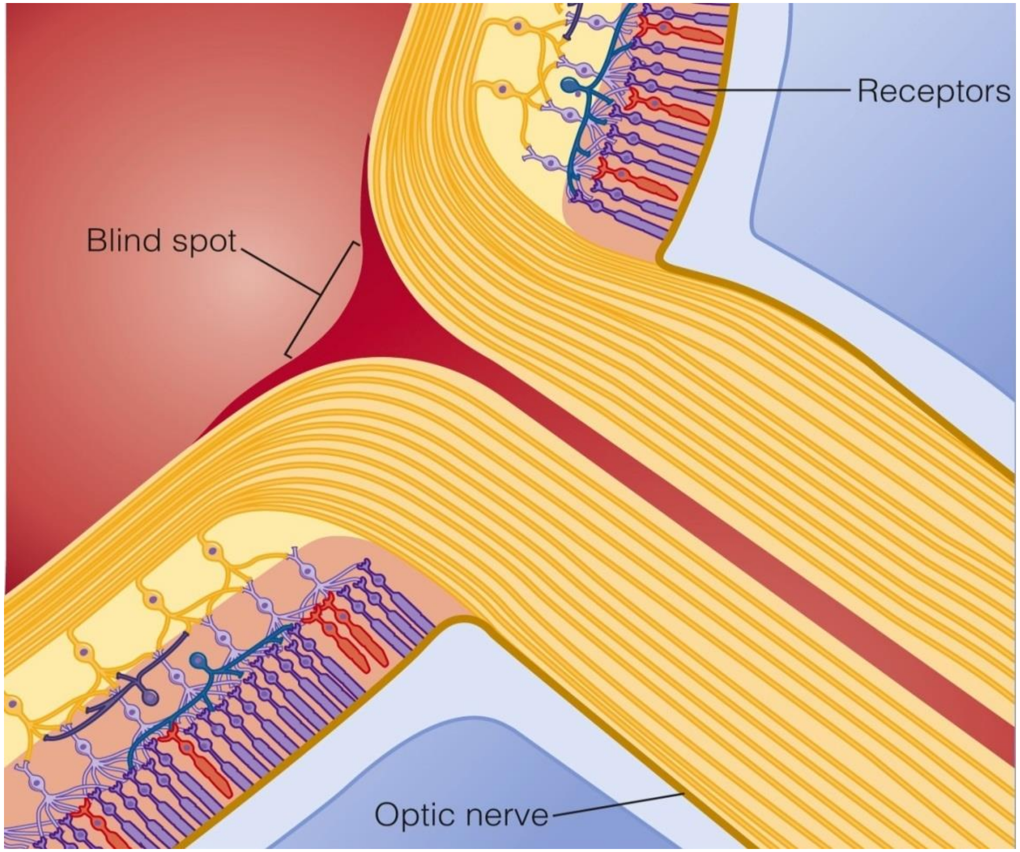
Vision – the retina

- Distribution of rods and cones across the retina



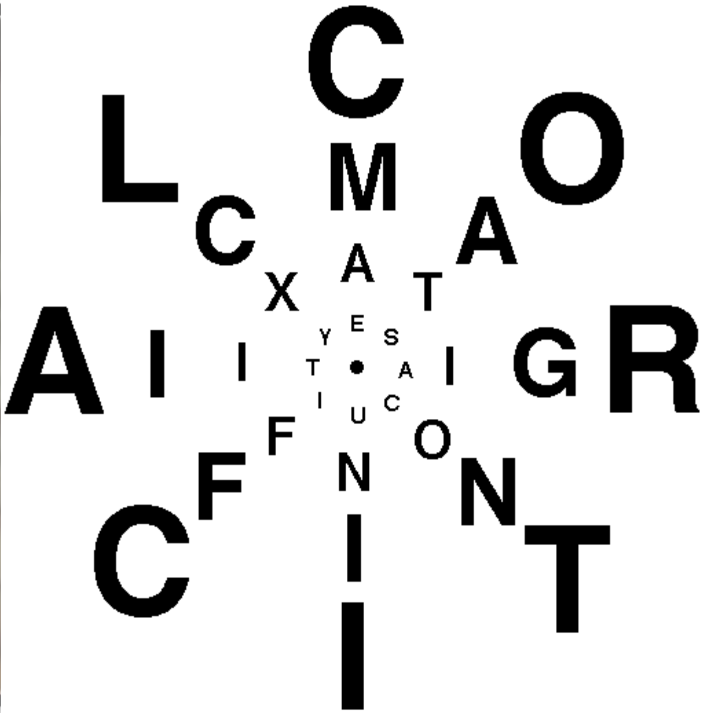
Vision – the retina

- Distribution of rods and cones across the retina



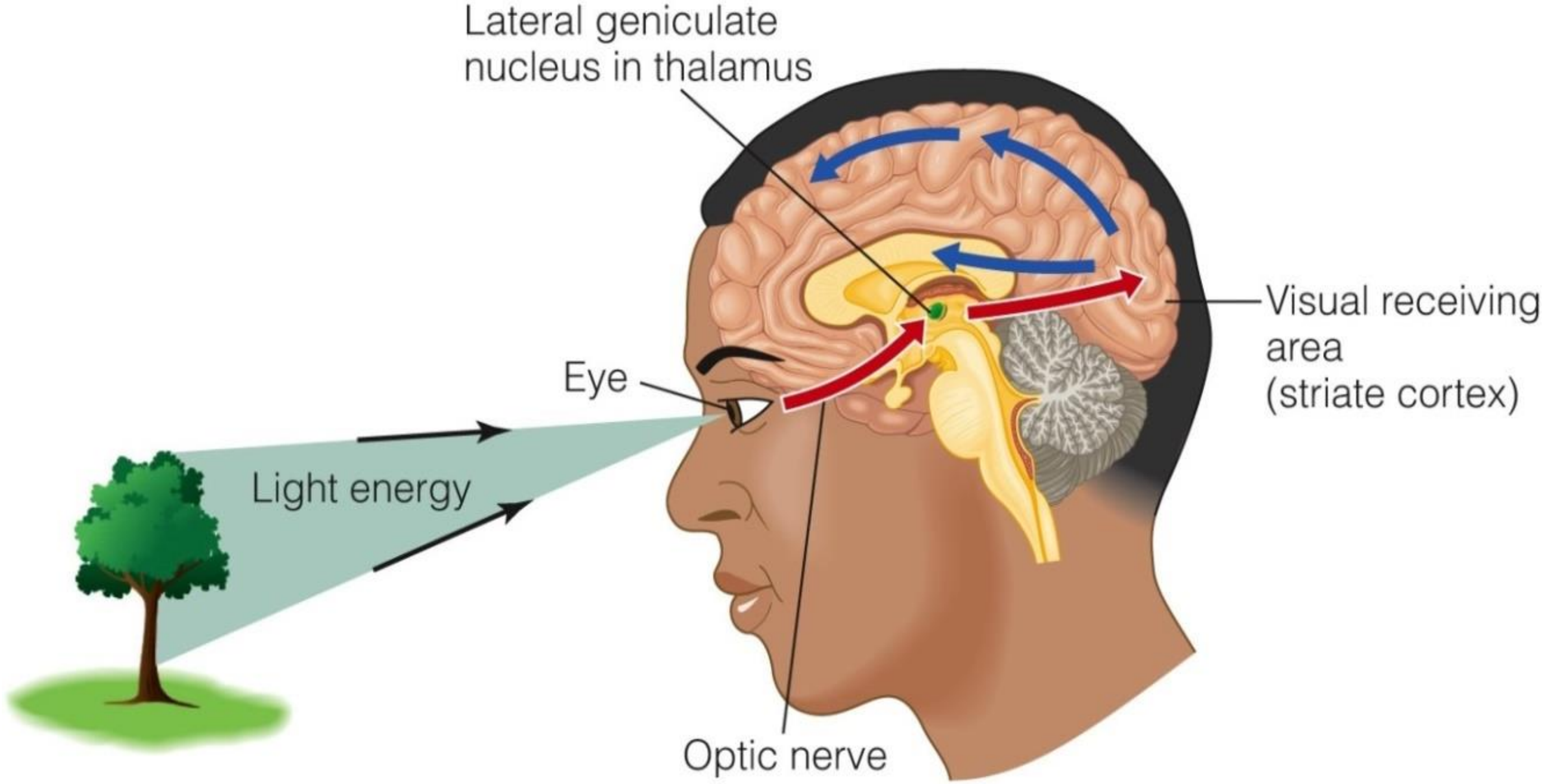
Vision – the retina

- Distribution of rods and cones across the retina
  - -> Higher resolution in the fovea



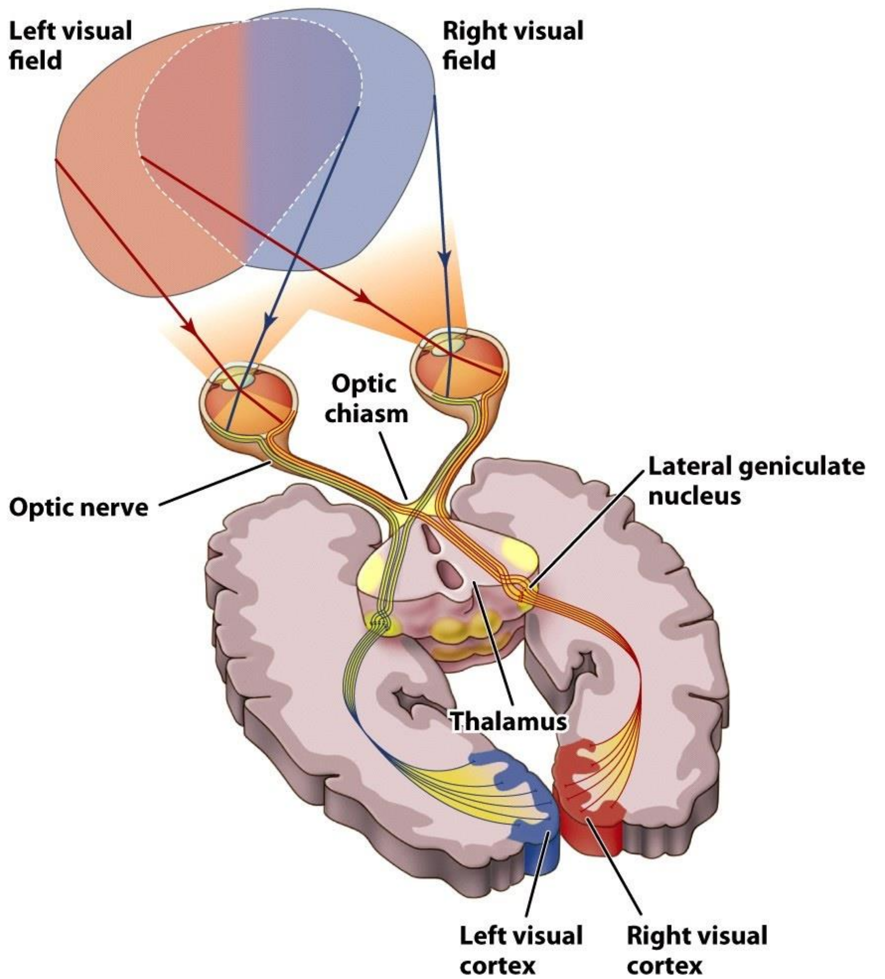
Vision – beyond the retina

- What happens after the retina?

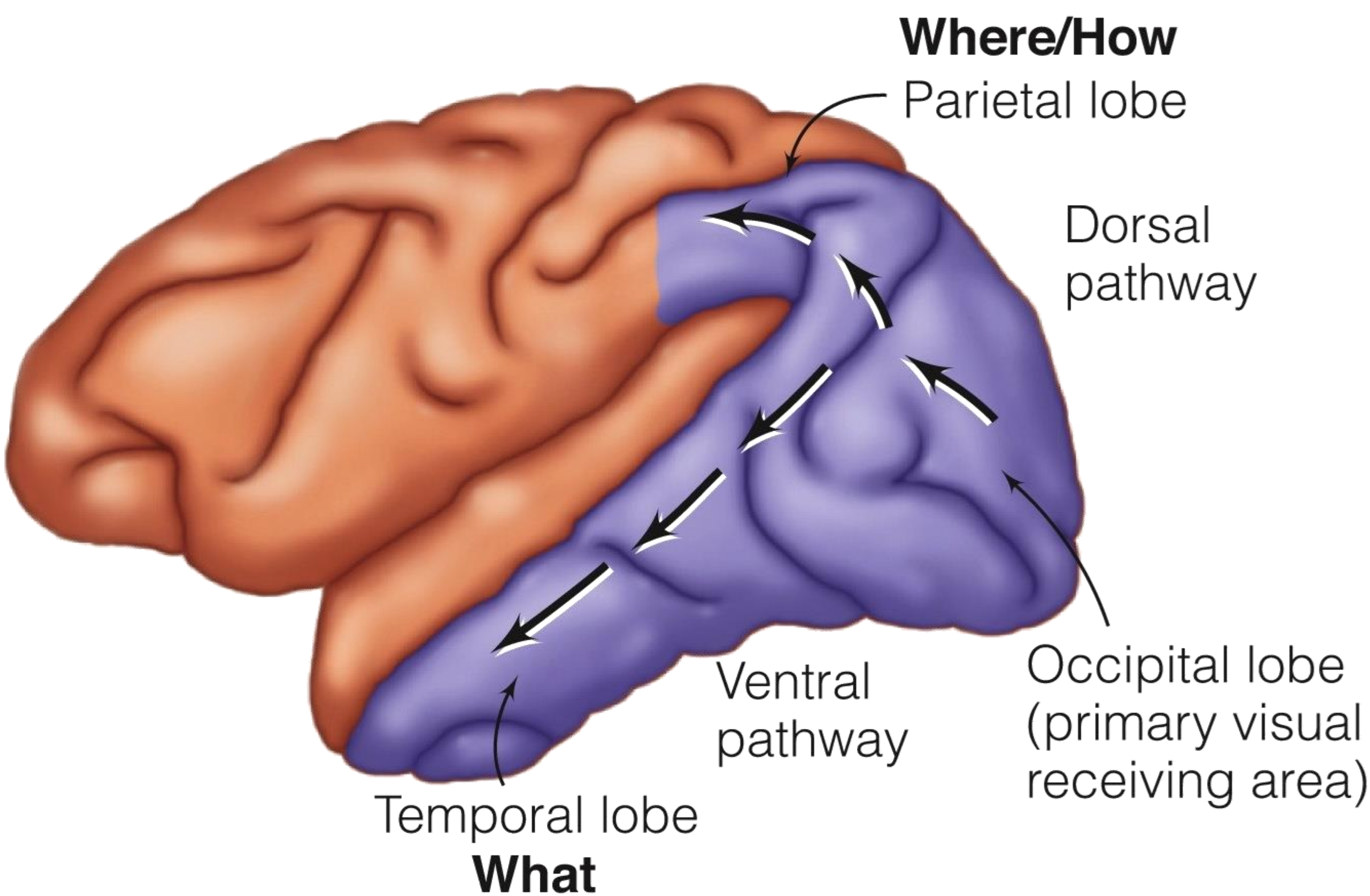


Vision – beyond the retina

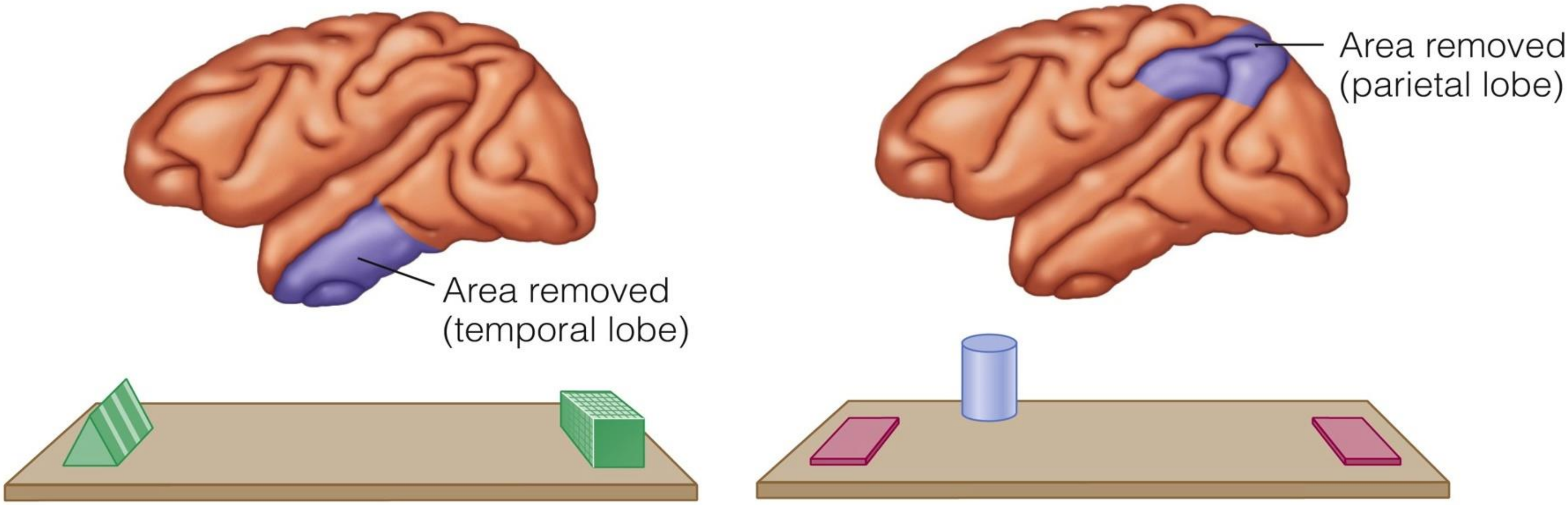
- What happens after the retina?



Vision – beyond the retina



Vision – beyond the retina



(a) Object discrimination

(b) Landmark discrimination

1982 Mishkin

Vision – beyond the retina

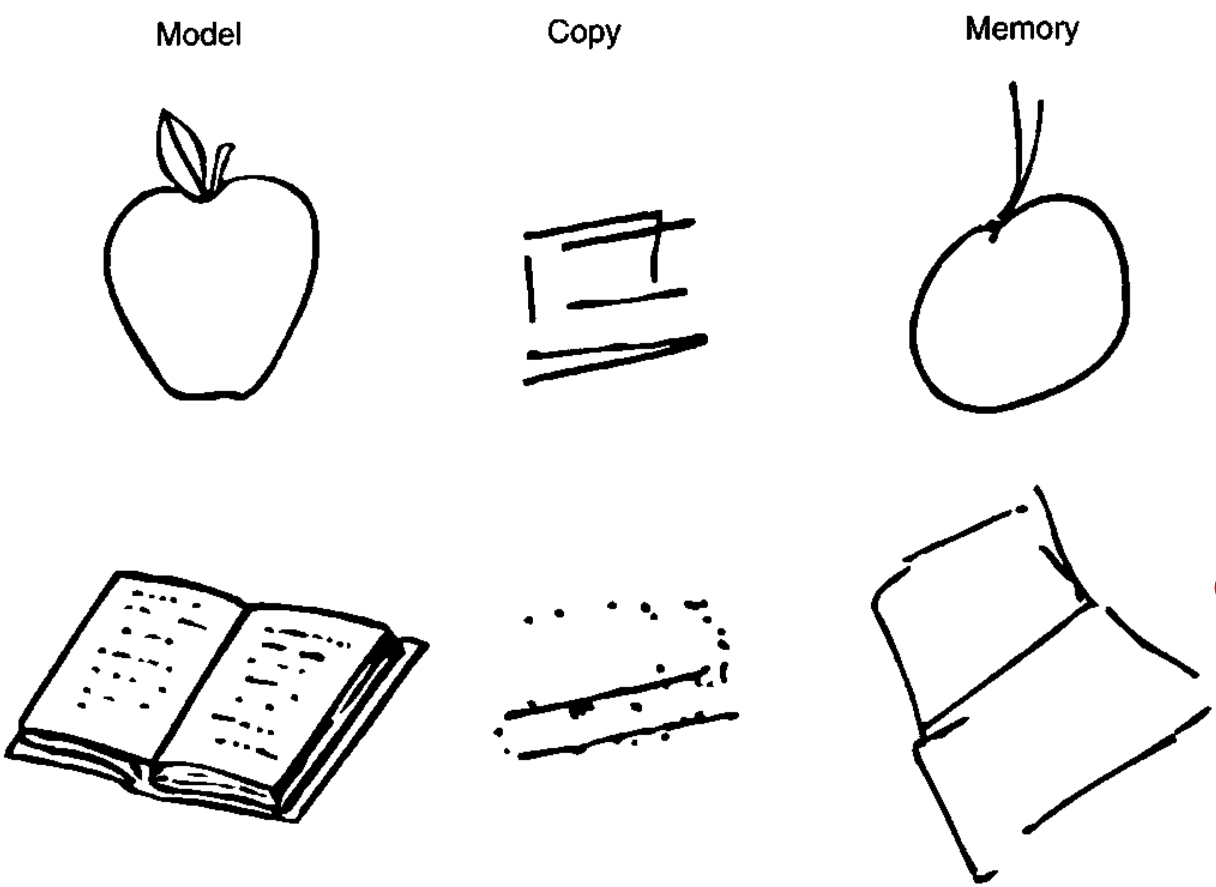
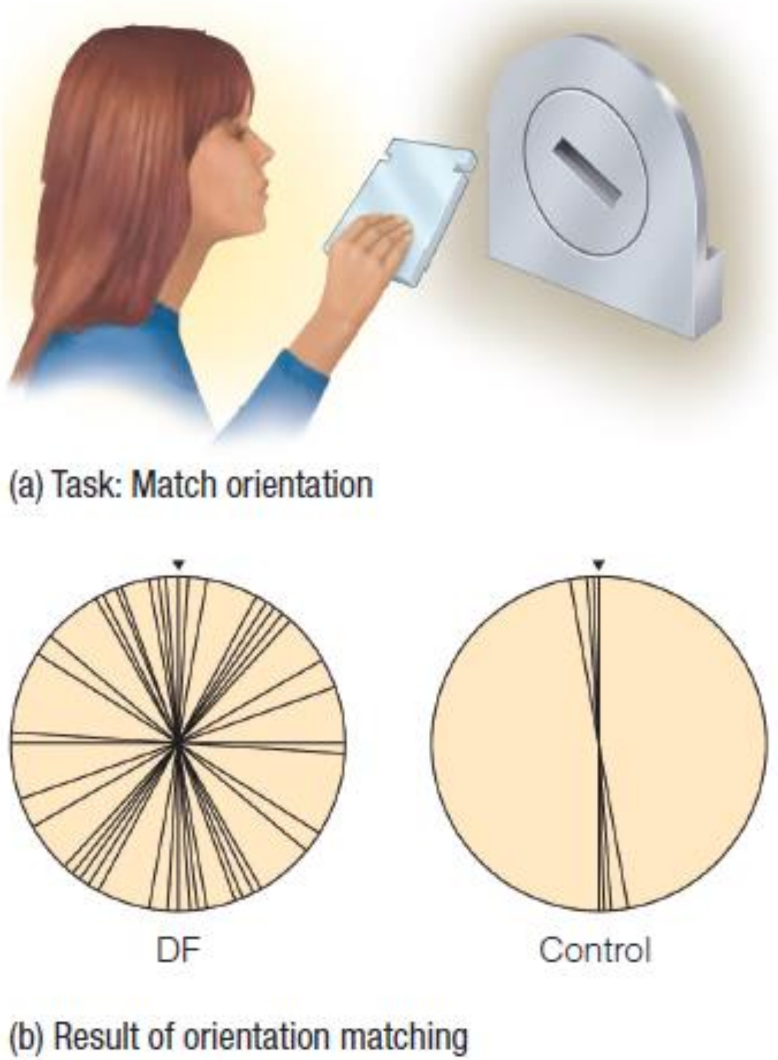
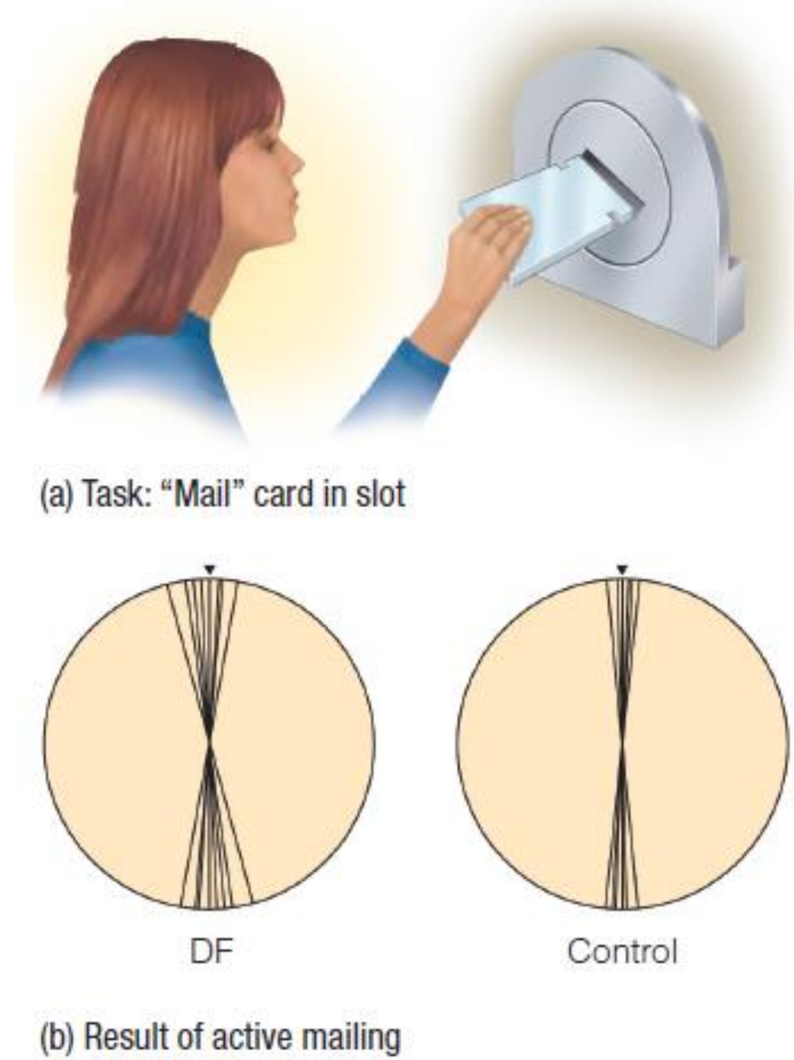


Figure 4.6
Patient D.F. could not recognize the apple or book drawings on the left and could not copy the drawings, as shown by her attempts in the middle column. The drawings on the right are ones she produced from memory. (From Milner & Goodale, 1995.)

Vision – beyond the retina



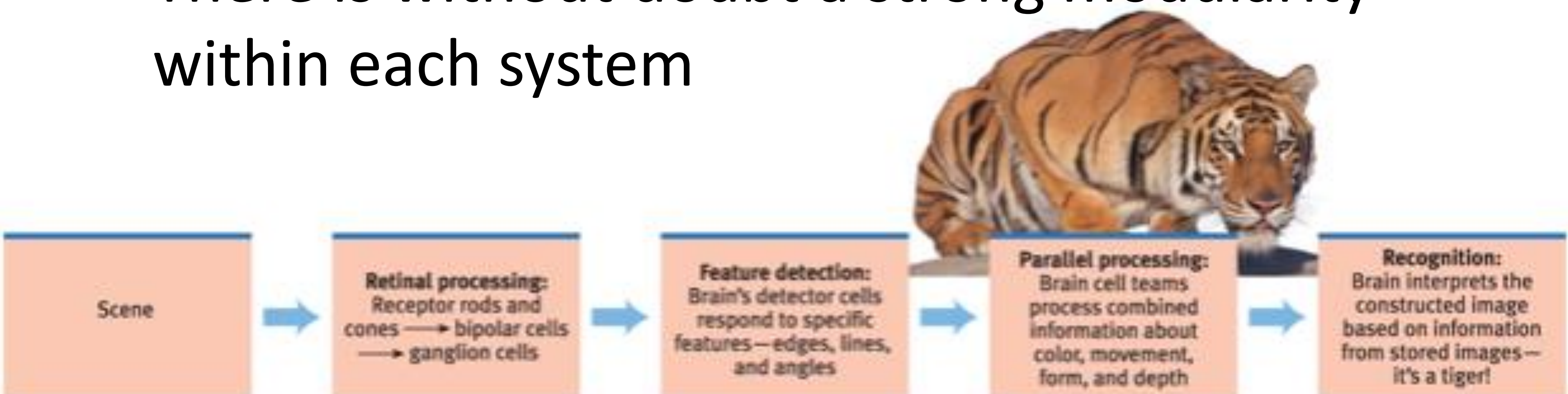


Milner & Goodale, 1995

Vision – beyond the retina

- Two systems
  - “where” (path to parietal cortex, dorsal)
  - “what” (path to temporal cortex, ventral)

- There is without doubt a strong modularity within each system



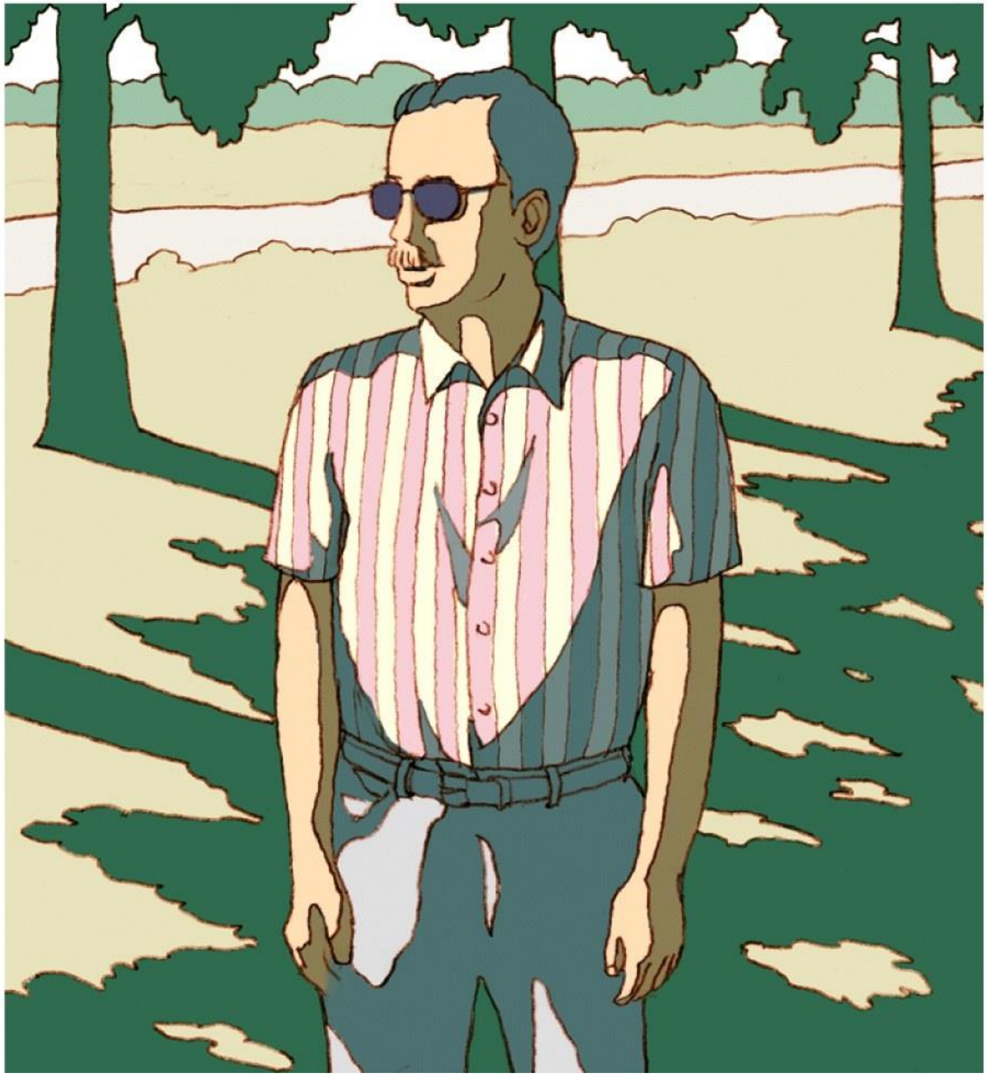
Myers/DeWall, *Psychology*, 13e, © 2021 Worth Publishers
Tom Walker/Getty Images

FIGURE 6.20 A simplified summary of visual information processing

Visual perception

The visual system is not a passive receiver of visual information but modulates, organizes, and interprets this information!

Visual perception



Visual perception

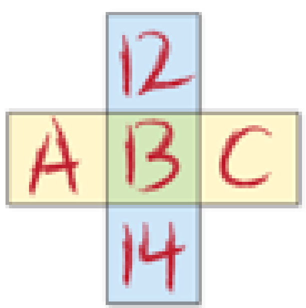
Visual perception

- Perception occurs on the basis of
  - Bottom-up (data-driven) processes
    - Starts at your sensory receptors and works up to higher levels of processing.
  - Top-down (knowledge-driven) processes
    - e.g., **priming**

A B C

Visual perception

- Perception occurs on the basis of
  - Bottom-up (data-driven) processes
    - Starts at your sensory receptors and works up to higher levels of processing.
  - Top-down (knowledge-driven) processes
    - e.g., priming
    - e.g., context



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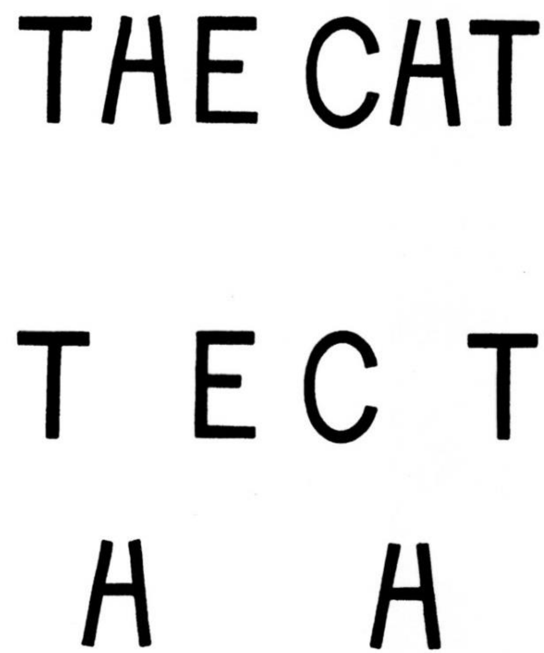


Context

- Our immediate context, and the motivation and emotion we bring to a situation, also affect our interpretations.

- People's expectations influence their perceptions constantly, whether about an angry man
  - Is he reaching for his keys or a weapon?

- "__eel is on the wagon."
  - Wheel
- "__eel is on the orange"
  - Peel

- How is the woman in the picture feeling?

Craig Klomparens/Hope College

FIGURE 6.9 Context makes clearer The Hope College volleyball team celebrates its national championship winning moment.

- Emotions can shove our perceptions in one direction or another
  - Hearing sad music can predispose people to perceive a sad meaning in spoken homophonic word
    - mourning – morning
    - die - dye
    - pain – pane

Visual perception

Visual perception

- Perception occurs on the basis of
  - Bottom-up (data-driven) processes
    - Starts at your sensory receptors and works up to higher levels of processing.
  - Top-down (knowledge-driven) processes
    - e.g., priming
    - e.g., context
    - Constructs perceptions from this sensory input by drawing on your experience and expectations.

https://en.wikipedia.org/wiki/French_fries

https://www.mcdonalds.com/us/en-us/product/small-french-fries.html

(Robinson et al., 2007)

(Lee et al., 2006)

What determines our perceptual experience?

- Experience →concepts, schemas
  - Helps organize and interpret unfamiliar information
  - Apply top-down processing to interpret ambiguous sensations

David

Diana

(Stern & Karraker, 1989)

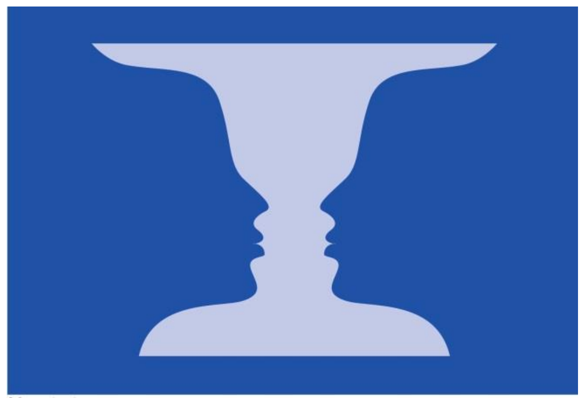
- Perception is influenced by culture.

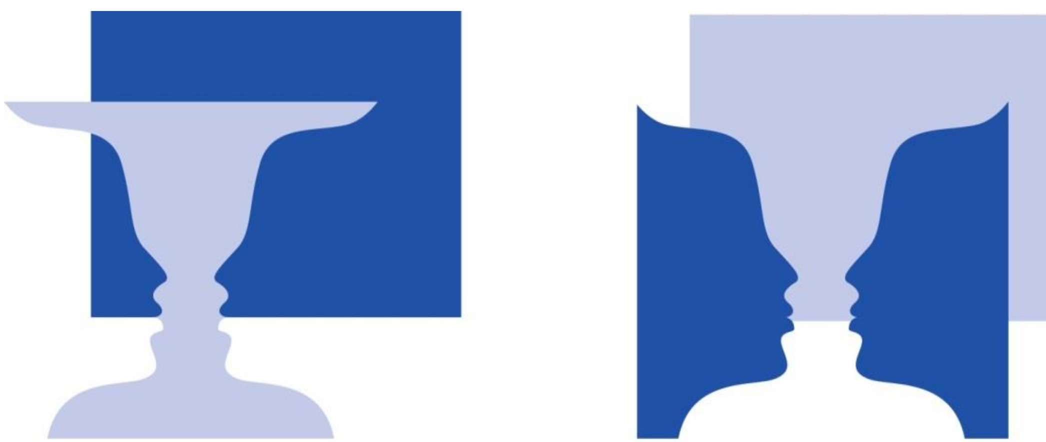
Myers/DeWall, *Psychology*, 13e, © 2021 Worth Publishers

(Gregory & Gombrich, 1973)

Perceptual organization

- Segregation of figure and ground

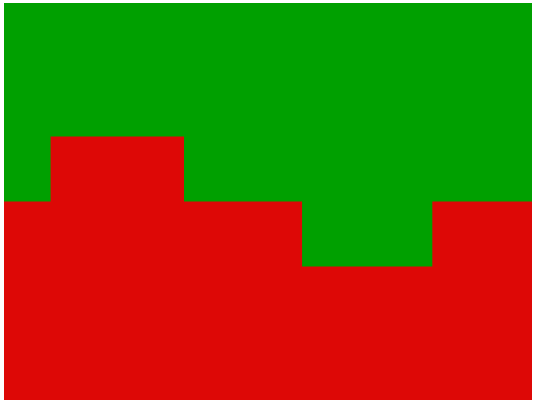




(a) (b)

Perceptual organization

- Principles of figure-ground segregation
  - Lower areas are more often perceived as figure than higher areas

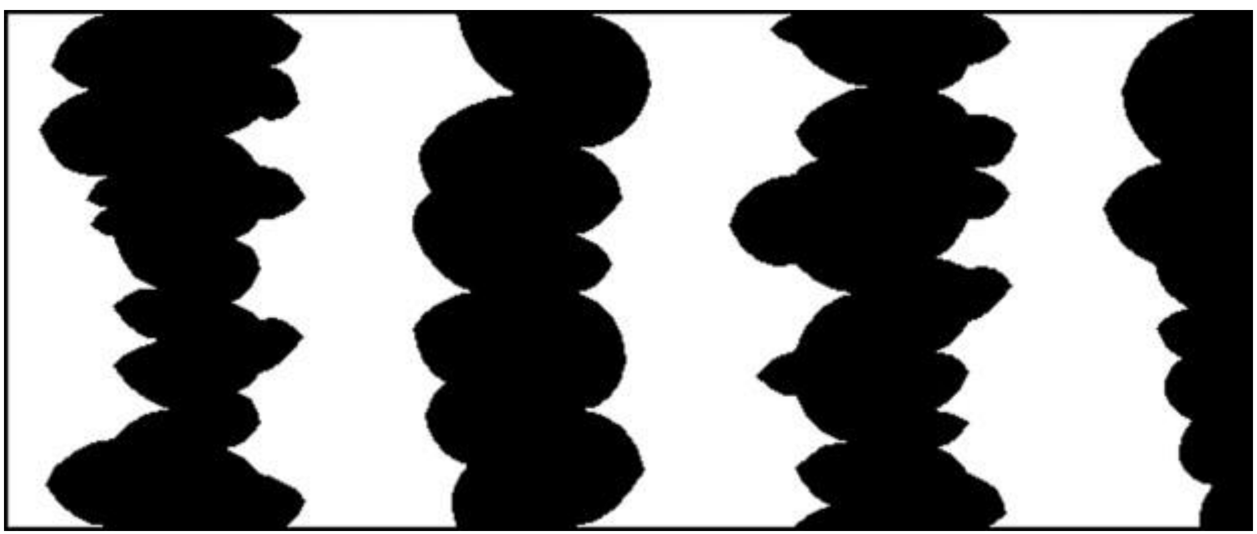


Perceptual organization

- Principles of figure-ground segregation
  - Lower areas are more often perceived as figure than higher areas

Perceptual organization

- Principles of figure-ground segregation
  - Lower areas are more often perceived as figure than higher areas
  - Convex forms are more often perceived as figure whereas concave forms are more often perceived as background

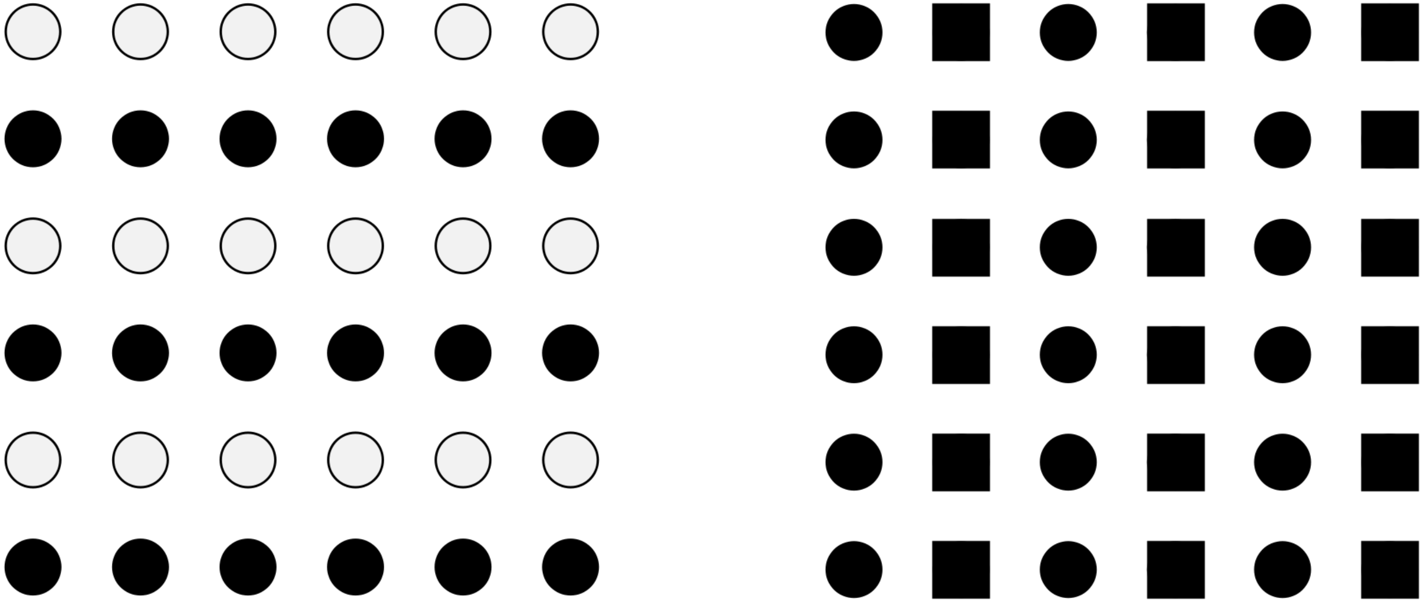


Perceptual organization

- Perceptual interpretation on the basis of organizing principles (Gestalt principles)
  - e.g., proximity

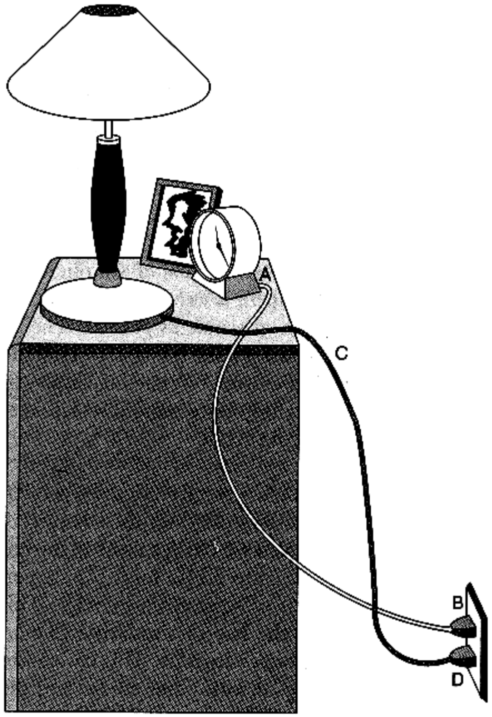
Perceptual organization

- Perceptual interpretation on the basis of organizing principles (Gestalt principles)
  - e.g., similarity



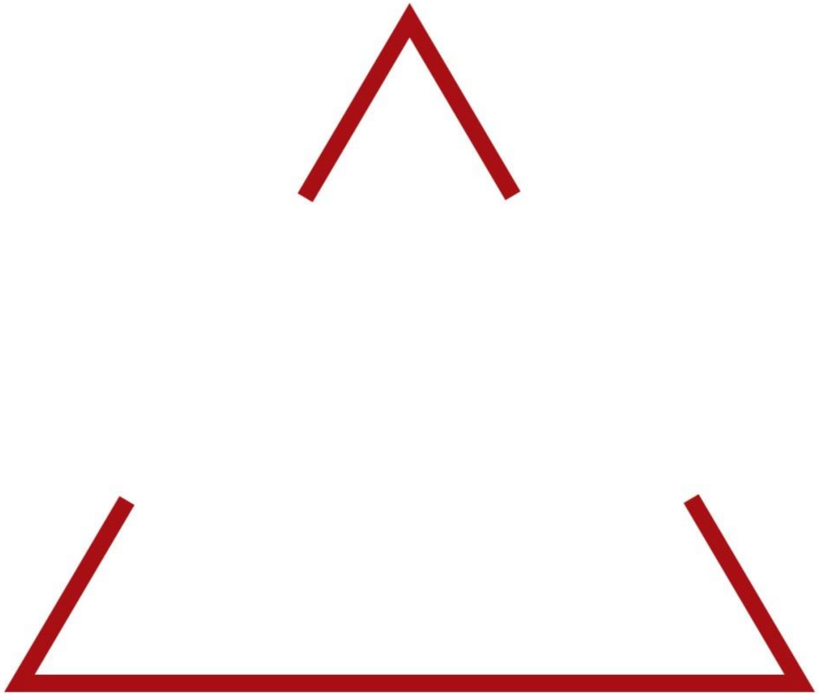
Perceptual organization

- Perceptual interpretation on the basis of organizing principles (Gestalt principles)
  - e.g., continuity



Perceptual organization

- Perceptual interpretation on the basis of organizing principles (Gestalt principles)
  - e.g., closure



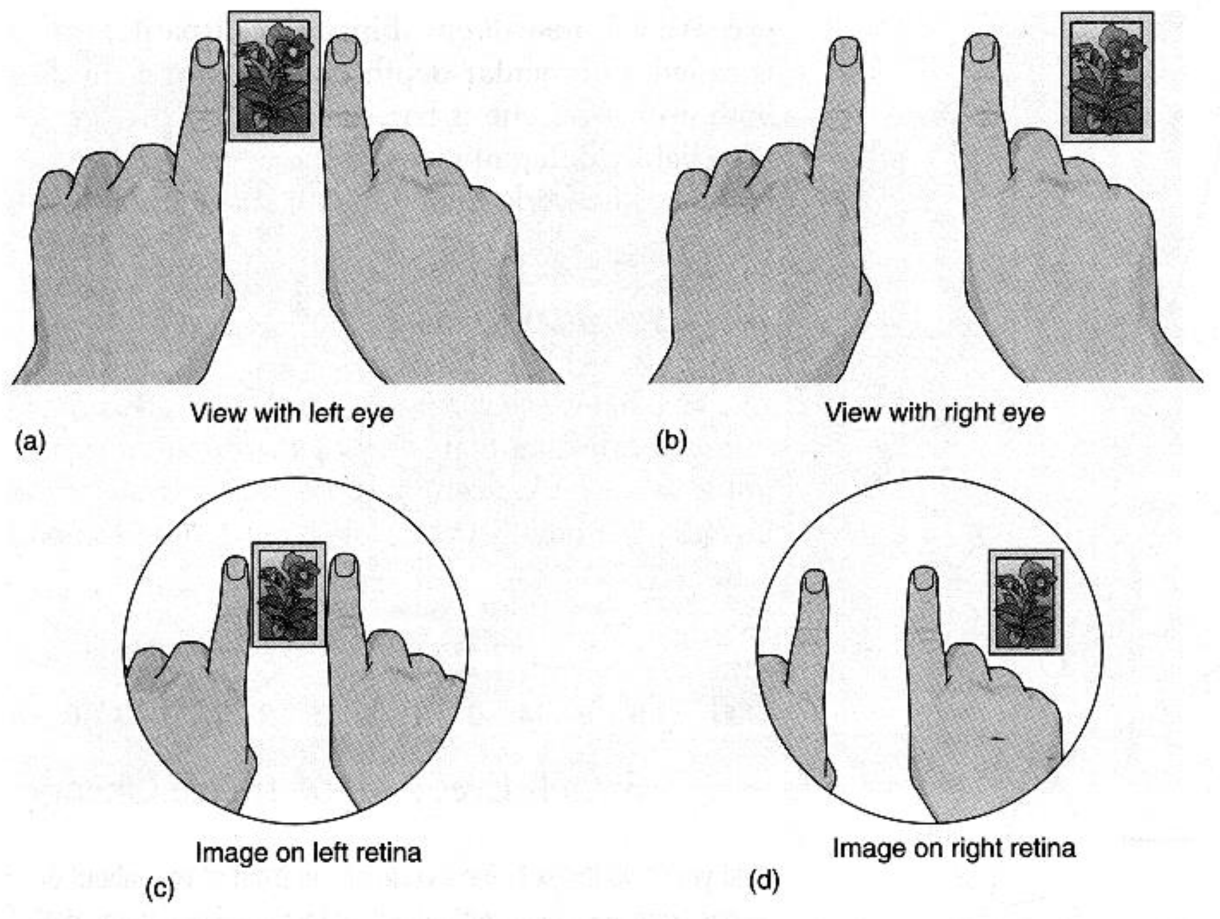
Perception of depth

- Depth cues
  - Binocular cues
    - Binocular disparity
    - Convergence

Monika Skolimowska/AP Images

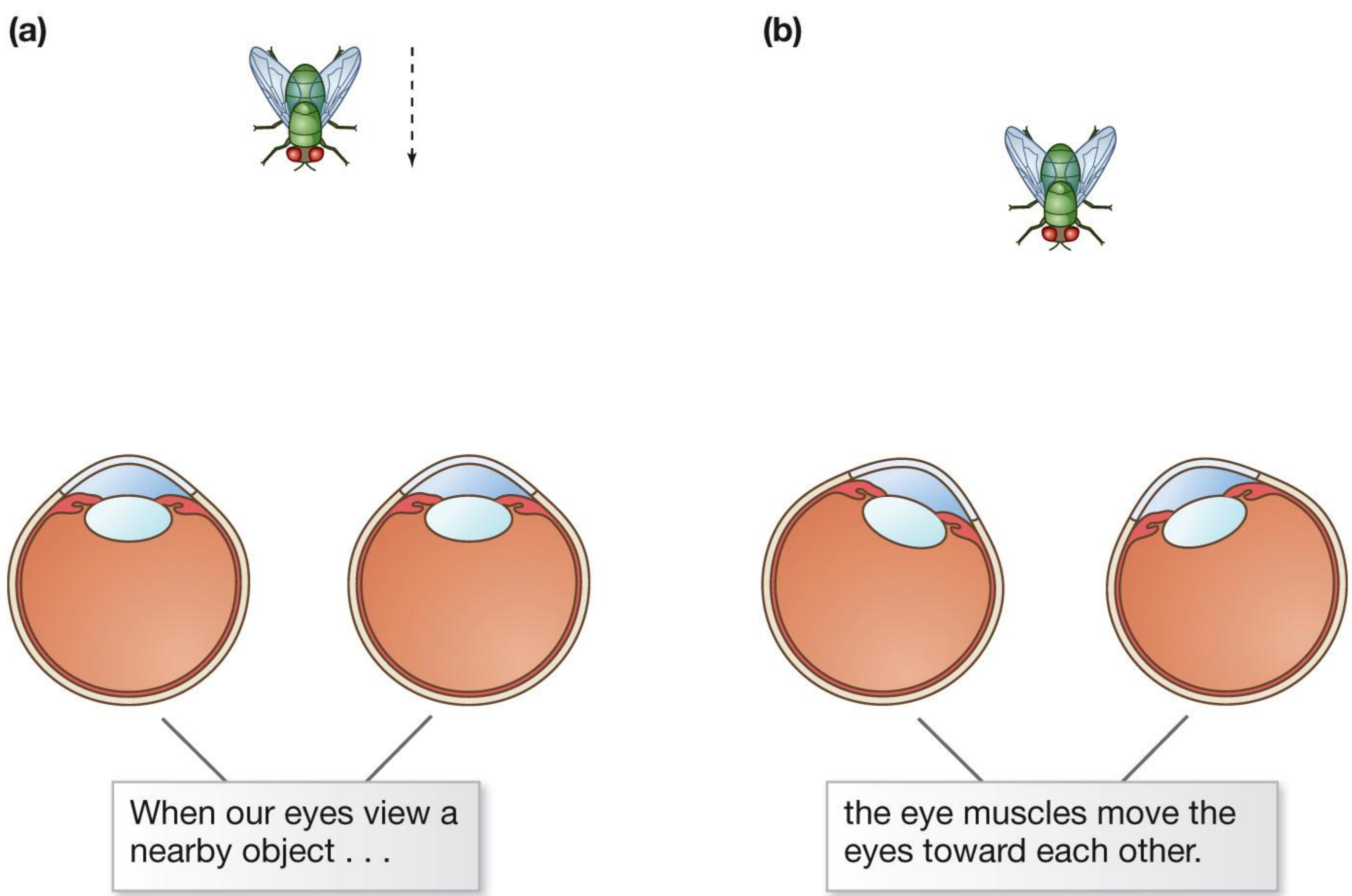
Perception of depth

- Binocular disparity



Perception of depth

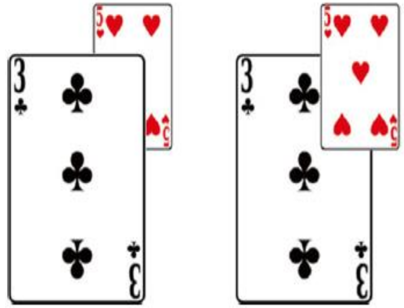
- Convergence



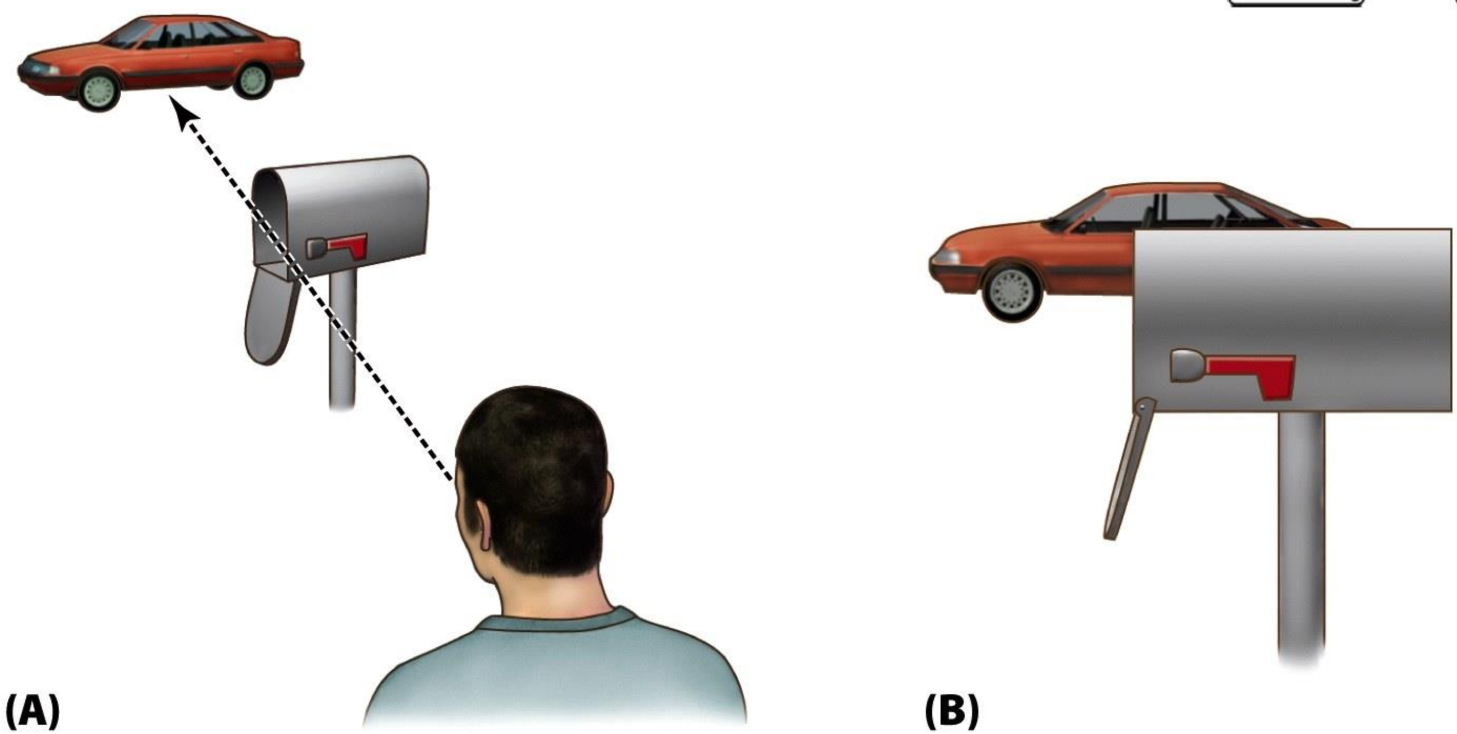
Perception of depth

- Depth cues
  - Binocular cues
    - Binocular disparity
    - Convergence
  - Monocular cues, e.g.,
    - Occlusion
    - Linear perspective
    - Texture gradients
    - Shadow
    - Motion parallax

Perception of depth



- Occlusion



Perception of depth

- Linear perspective

Perception of depth

- Texture gradients

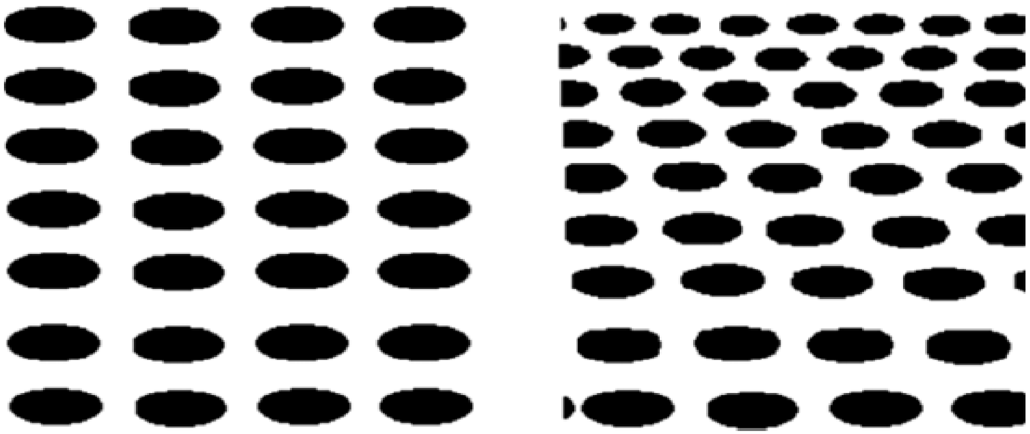
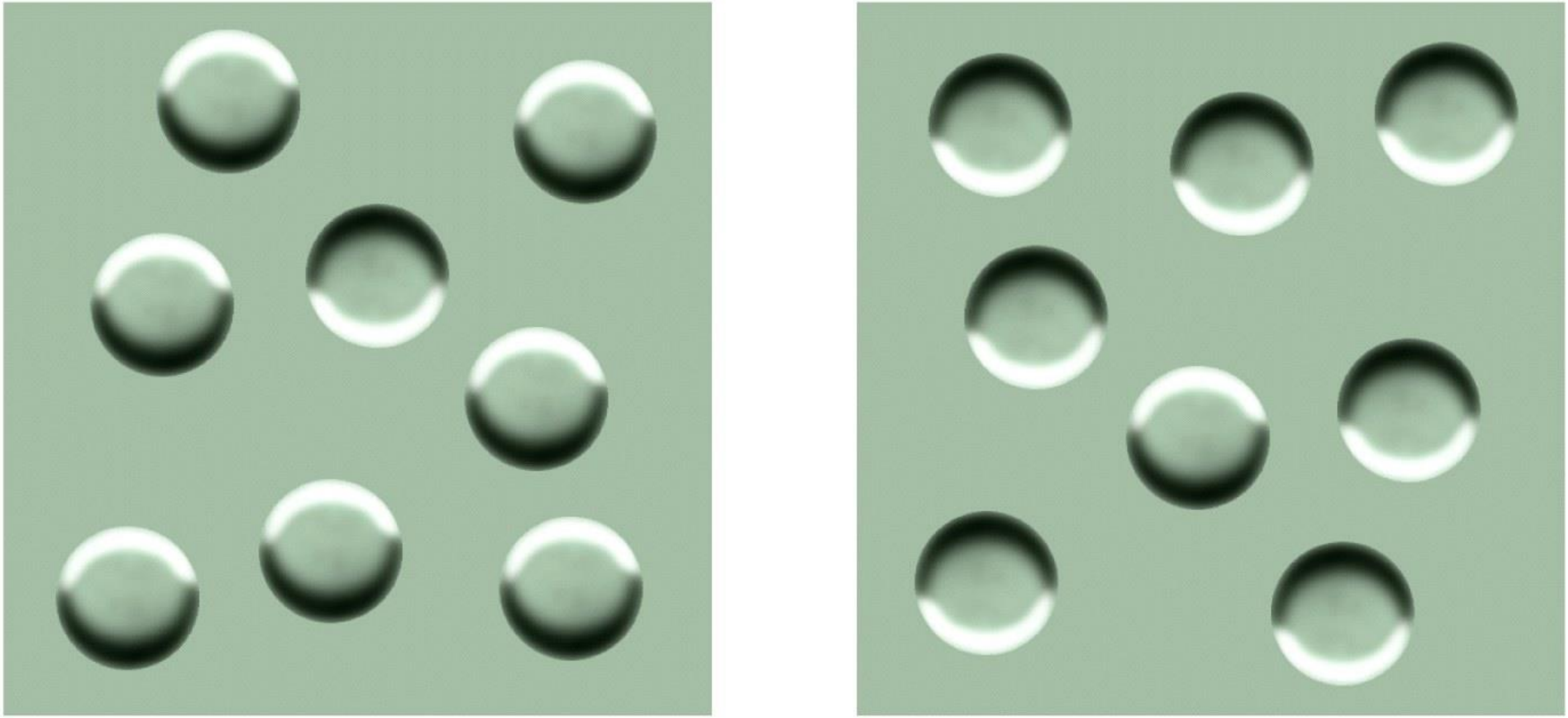


Fig. 7.2 When texture gradients are uniform, as on the *left*, no impression of depth is created; however, compressing points and using heterogeneous point size, on the *right*, give an impression of depth

Perception of depth

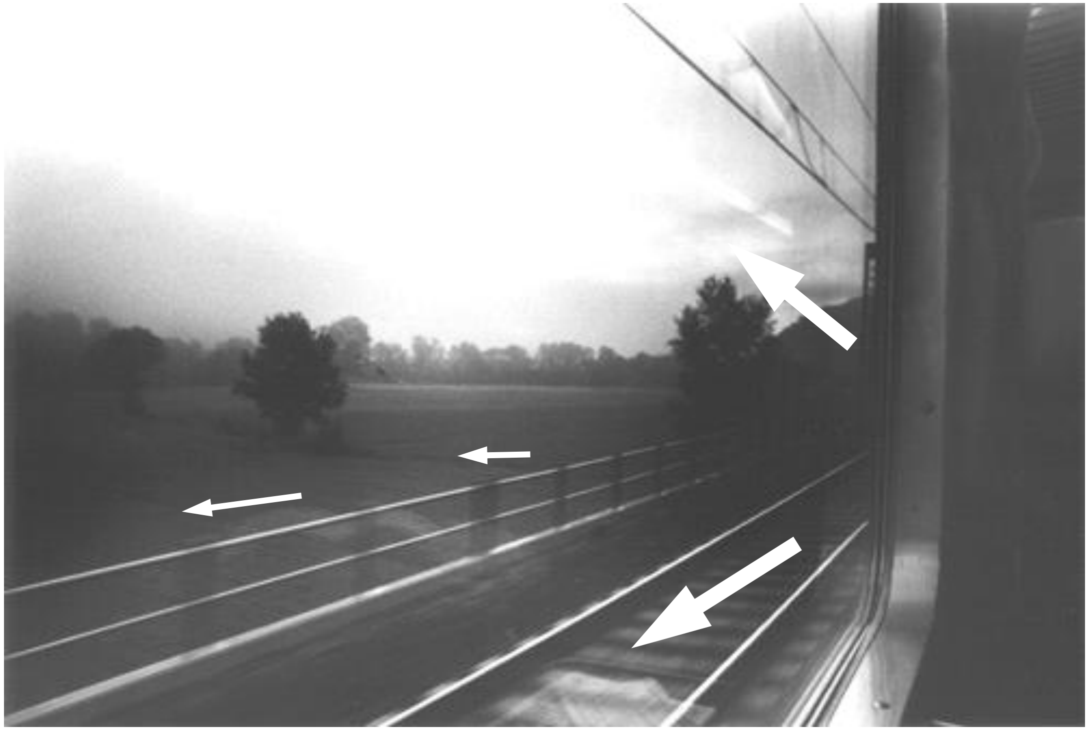
- Shadow



(A) (B)

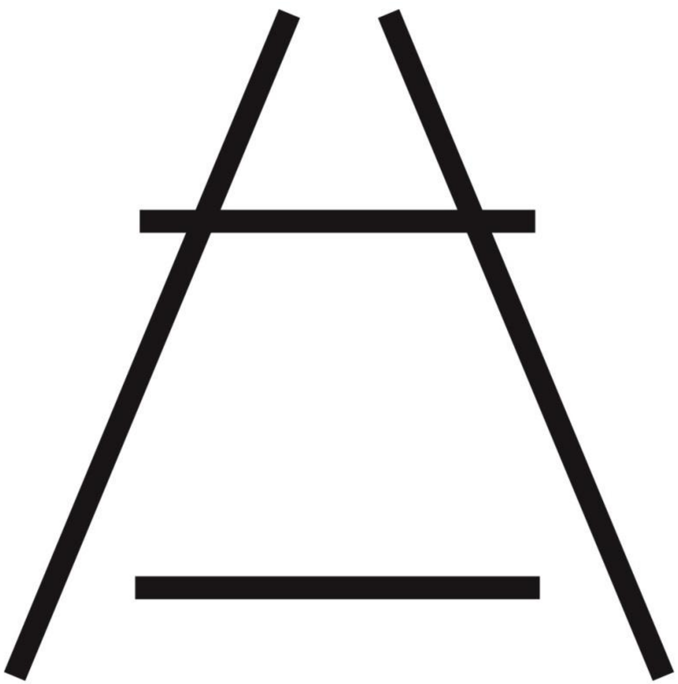
Perception of depth

- Motion parallax



Perception of size

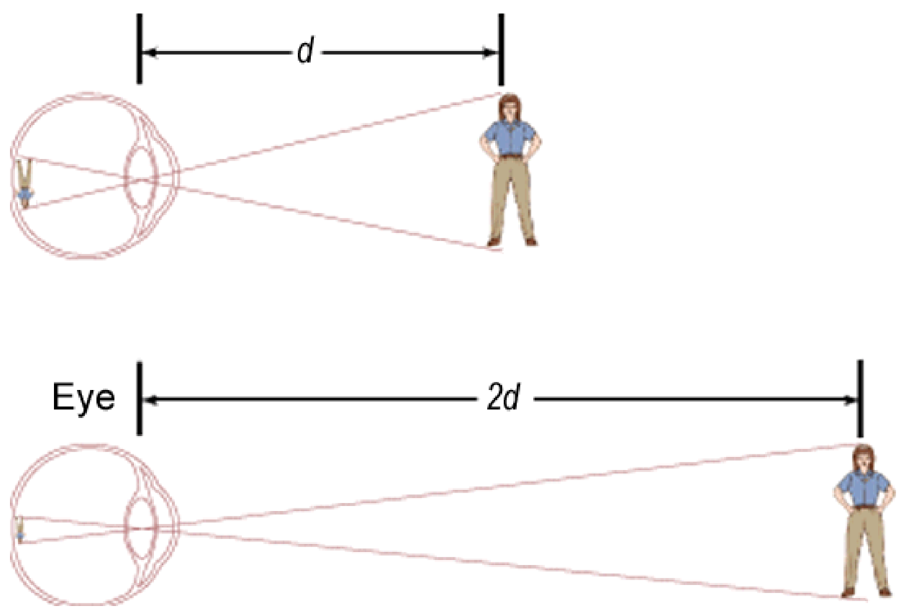
- Size perception depends on perceived distance

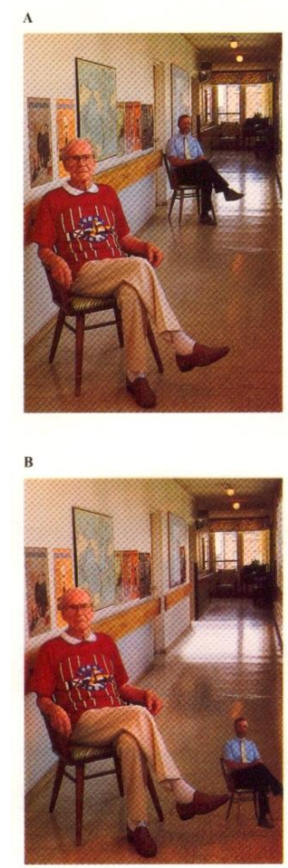


"seeing is not just a simple stimulation of retinal cells"

Perceptual constancies

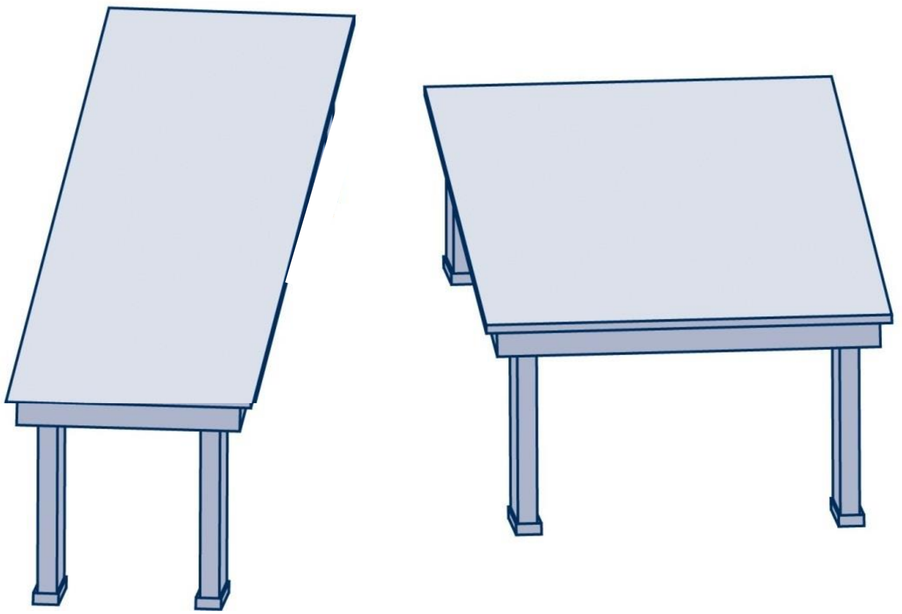
- We correctly perceive different properties of distal stimuli despite continuously changing proximal stimuli.
  - Size constancy





Perceptual constancies

- We correctly perceive different properties of distal stimuli despite continuously changing proximal stimuli.
    - Size constancy
    - Shape constancy

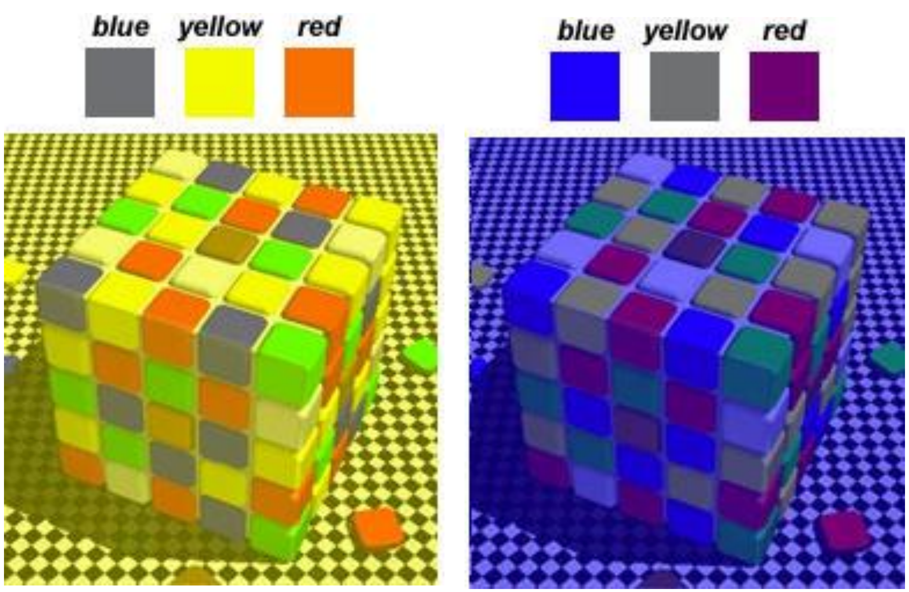


Perceptual constancies

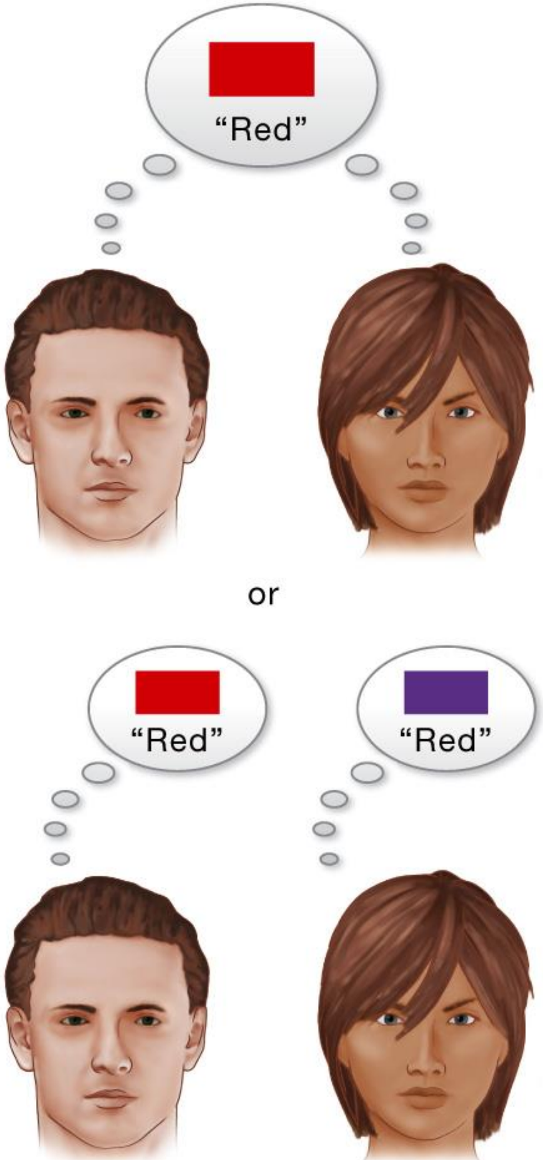
- We correctly perceive different properties of distal stimuli despite continuously changing proximal stimuli.
  - Size constancy
  - Shape constancy
  - Lightness constancy

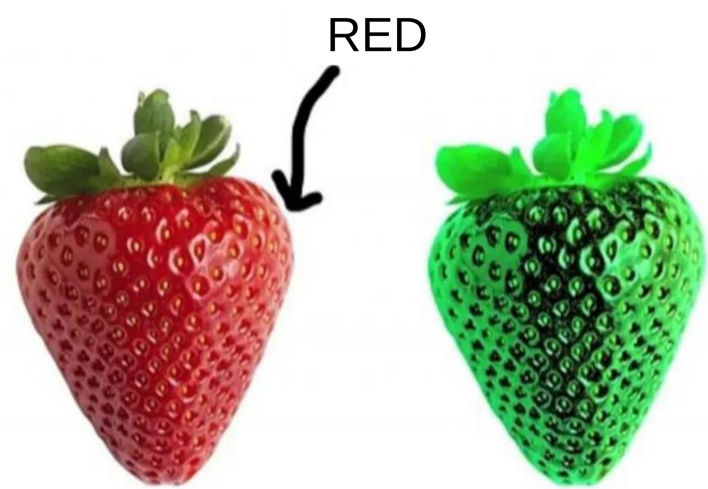
Perceptual constancies

- We correctly perceive different properties of distal stimuli despite continuously changing proximal stimuli.
  - Size constancy
  - Shape constancy
  - Lightness constancy
  - Color constancy



Qualia



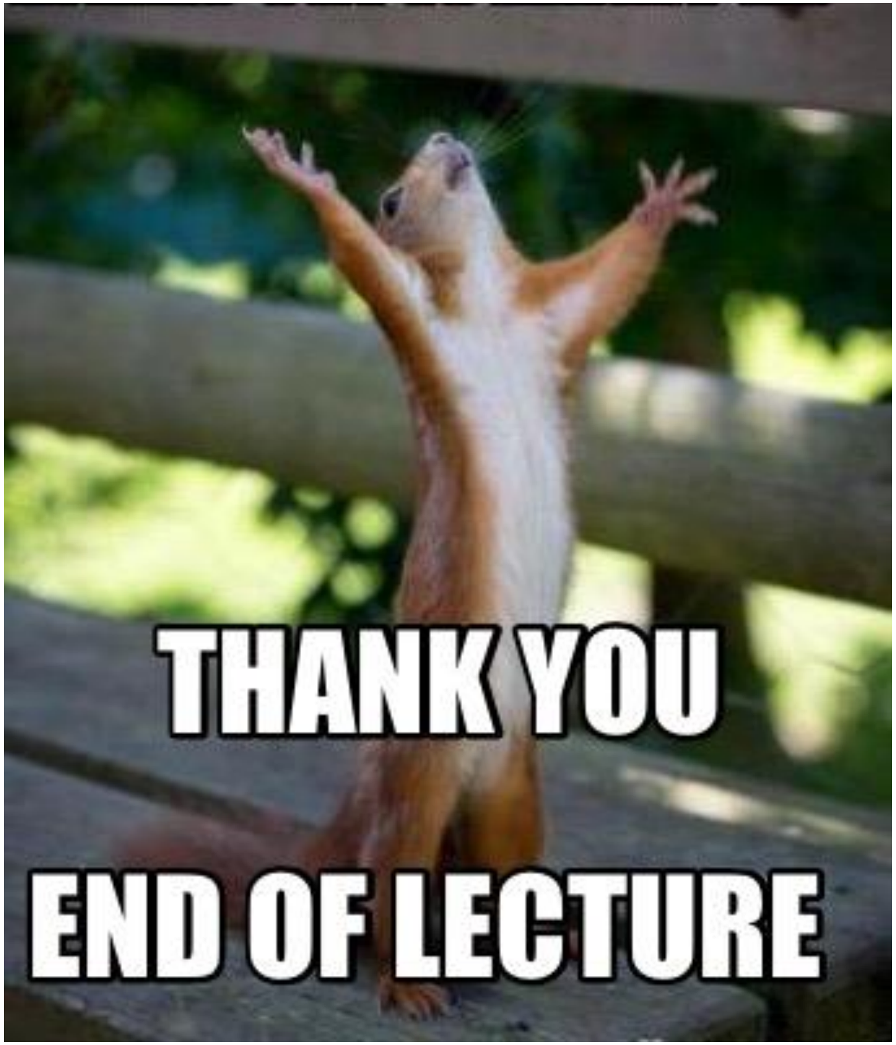


https://evrimagaci.org/kualia-ve-renkler-herkes-renkleri-ayni-mi-gorur-3755

Predictions (bonus content)

https://www.cnet.com/science/biology/features/1-puzzling-reason-ai-may-never-compete-with-human-consciousness/

https://slate.com/technology/2017/04/heres-why-people-saw-the-dress-differently.html


THANK YOU
END OF LECTURE